

Review

Groundwater Potential Mapping Using Machine Learning Techniques: Current Trends and Future Perspectives

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Abstract

Groundwater is a vital freshwater resource that supports domestic, agricultural, and industrial activities in many regions worldwide. Accurate groundwater potential mapping (GPM) is essential for sustainable water resource management; however, traditional empirical and statistical approaches often struggle to capture the complex, nonlinear relationships among hydrogeological variables. In recent years, machine learning (ML) has emerged as a powerful data-driven approach for improving GPM accuracy and efficiency. This review synthesizes findings from 83 peer-reviewed studies published between 2015 and 2025, focusing on widely used ML algorithms such as Random Forest, Support Vector Machines, Artificial Neural Networks, and hybrid models. The review evaluates key methodological aspects, including input parameter selection, data partitioning, integration with GIS and remote sensing, and model justification techniques. It also discusses common challenges such as data limitations, regional variability, and model interpretability. The results indicate that ML-based approaches can significantly enhance groundwater prediction when supported by appropriate data and validation strategies. Future research directions include explainable artificial intelligence, uncertainty quantification, multi-source data integration, and improved model transferability. This review provides a comprehensive reference for advancing reliable and sustainable groundwater potential mapping.

Keywords: Hydrogeology; machine learning; geographic information system (GIS); remote sensing (RS); artificial intelligence (AI); groundwater resources; explainable AI (XAI)



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1. Introduction

Water is a finite and vital resource for preserving ecological balance and supporting human [1]. As global water demand continues to rise due to rapid population growth, socioeconomic development [2,3], and intensified climate change impacts [4–8], groundwater emerges as an essential component of the global freshwater supply, playing a critical role in ensuring water security for agricultural, domestic, and industrial needs [9–12]. It sustains

over 40% of global irrigated agriculture, while also providing drinking water for nearly half of the world's population [13]. Unlike surface water, groundwater is generally more resilient to seasonal variability and drought conditions, making it an essential buffer against environmental stress and water scarcity, particularly in arid and semi-arid regions [14,15]. In many developing countries, groundwater is often the only reliable source of clean water, sustaining both rural communities and rapidly growing urban centers [9,16]. Despite its importance, groundwater resources are increasingly threatened by overexploitation, contamination, and poor management practices. Therefore, accurate identification and mapping of groundwater potential zones are fundamental to safeguarding this invisible resource and ensuring long-term water security.

Groundwater potential (GWP) refers to the likelihood of groundwater occurrence, recharge, or yield within a given hydrogeological setting, as controlled by a combination of geological, hydrological, topographical, and climatic factors [17]. Building upon this concept, GPM has been widely developed as a spatial approach to delineate zones with varying groundwater availability using integrated geospatial and statistical techniques [18]. Early studies and subsequent reviews (e.g., Díaz-Alcaide et al. [19] and references therein) have established that GPM relies on the systematic integration of multiple conditioning factors to estimate groundwater occurrence and productivity. GPM is typically conducted through a systematic, multi-stage workflow that integrates hydrogeological knowledge, spatial analysis, and predictive modeling. The process generally involves: (1) acquisition of relevant spatial datasets, including geological, hydrological, topographical, climatic, and remote sensing-derived variables; (2) preprocessing and thematic layer preparation within a Geographic Information System (GIS) environment; (3) selection and normalization of groundwater conditioning factors using statistical techniques; (4) model development using statistical approaches; (5) justification and performance assessment through metrics such as ROC-AUC, accuracy, cross-validation, and sensitivity analysis; and (6) generation of groundwater potential index maps followed by classification into suitability zones (e.g., very low to very high potential) [19,20]. Traditionally, these indicators were combined according to their relative influence on groundwater potential using mathematical and statistical approaches, such as the Frequency Ratio (FR), Weights-of-Evidence (WoE), and multi-criteria decision analysis (MCDA) methods like the Analytical Hierarchy Process (AHP), which have been widely used to delineate areas with high groundwater availability [21–24]. While these methods are simple and require relatively low computational resources, they are often limited by their reliance on subjective expert judgment, fixed weighting schemes, and the assumption of linear relationships among influencing factors. These limitations reduce their adaptability to spatial heterogeneity and complex, non-linear hydrogeological interactions. Additionally, conventional methods may not fully utilize the growing availability of large-scale spatial datasets derived from remote sensing (RS) and GIS platforms. As a result, traditional techniques may yield less accurate or overly generalized results, making them insufficient for capturing the dynamic and multifactorial nature of groundwater systems in diverse geological settings.

Recently, artificial intelligence (AI) has exhibited rapid expansion, especially through machine learning (ML) models [25,26], due to their ability to model complex, nonlinear relationships among multiple interacting variables [27–29]. Unlike traditional groundwater assessment methods, ML algorithms can learn directly from data without relying on pre-defined equations or assumptions, making them particularly well-suited for handling the inherent uncertainty and spatial variability of subsurface systems [17,30]. The increasing availability of high-resolution geospatial, hydrological, and RS data has further accelerated the adoption of ML techniques in GPM, aquifer characterization, water quality prediction, and recharge estimation [31,32]. Models such as Random Forest (RF), Support Vector Ma-

chines (SVMs), Artificial Neural Networks (ANNs), and Gradient Boosting Machines (e.g., XGBoost) have demonstrated superior performance in predictive accuracy and robustness compared to conventional approaches [27,31,33]. Moreover, ML's capacity to integrate diverse data sources, such as topographic, geological, climatic, and land use, offers a more holistic and data-driven framework for informed groundwater management. This paradigm shift marks a transformative step toward more adaptive, scalable, and precise hydrogeological modeling. Despite remarkable advancements, several challenges persist in ML-based GPM. Data heterogeneity, spatial autocorrelation, feature redundancy, and overfitting remain critical issues affecting model transferability across diverse hydrogeological contexts. Furthermore, the interpretability of complex models, the selection of optimal conditioning parameters, and the limited accessibility of high-quality training data continue to hinder the operational application of ML in groundwater management. Addressing these challenges requires systematic analysis of existing approaches, transparent reporting of methodological workflows, and the development of generalizable, explainable, and scalable ML frameworks.

Therefore, this review aims to provide a comprehensive synthesis of the current state of ML applications in GPM, highlighting their methodological evolution, data integration strategies, and performance trends over the past decade (2015–2025). Specifically, it (1) summarizes key ML techniques and their comparative performance in GPM; (2) categorizes commonly used input parameters and data sources; (3) reviews methodological workflows including feature selection, model training, and validation; and (4) identifies emerging trends, challenges, and future research directions. By consolidating existing knowledge, this study provides an updated roadmap for researchers, practitioners, and policymakers aiming to advance the use of machine learning for sustainable groundwater exploration and management.

In this study, the terms training, validation, and testing are used with specific and consistent meanings. The training dataset is used to develop the model by learning the relationships between input variables and the target output. The validation dataset is employed during model development for hyperparameter tuning and model selection, often through cross-validation techniques. The testing dataset is reserved for the final evaluation of model performance and provides an unbiased estimate of predictive capability on unseen data. It is important to note that, in the groundwater potential mapping literature, the term “validation” is sometimes used interchangeably with “testing.” However, in this review, these terms are clearly distinguished to ensure methodological consistency and clarity.

2. Methodology

2.1. Literature Search and Selection

To systematically investigate the application of ML techniques in GPM, a structured review methodology was adopted. The Web of Science (WoS) database was selected as the primary source for literature retrieval due to its comprehensive indexing of high-quality, peer-reviewed journals across multidisciplinary scientific fields, including environmental sciences, hydrology, and geoinformatics. A combination of relevant keywords and Boolean operators was employed to search for articles published between January 2000 and June 2025. These search terms included phrases such as “Groundwater Potential,” “Groundwater Zonation,” “machine learning,” “Artificial Intelligence,” “Random Forest,” “Support Vector Machine,” and related terms. Only journal articles written in English were considered. After retrieving the initial set of publications, a relevance screening process was conducted by manually reviewing titles and abstracts to exclude studies that were outside the scope of GPM (e.g., those focused solely on groundwater quality, recharge estimation,

or hydrochemistry), did not employ ML techniques, or were duplicated entries. This was followed by a full-text review of the remaining articles to ensure that they provided sufficient methodological detail and met the inclusion criteria. A total of 140 studies were selected for in-depth analysis. Key information was extracted from each paper, including the year of publication, study area, ML models used, input variables, data sources, model tuning approaches, evaluation metrics, and the integration of geospatial tools such as GIS or remote sensing. This structured approach ensured a rigorous and reproducible review process, allowing for meaningful synthesis of current trends, research gaps, and future directions in the field. Figure 1 shows the schematic workflow of the review process for GPM studies using ML, showing the stages from database search to data analysis.

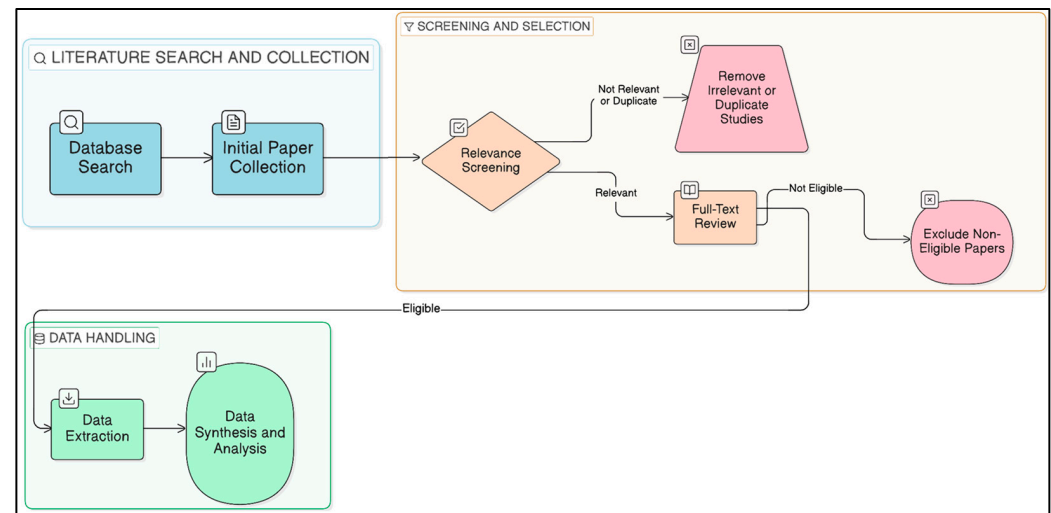


Figure 1. Schematic workflow of the review process.

2.2. Bibliometric Analysis

To enhance the clarity and analytical depth of the bibliometric analysis, a thesaurus file was employed in VOSviewer (version 1.6.20), a bibliometric analysis and visualization software, to unify semantically equivalent keywords under a single representative term. For instance, various expressions referring to the concept of groundwater potential, such as GPM, groundwater potentiality, groundwater potential zone, and groundwater management, were standardized to the unified term GPM. This consolidation not only minimized redundancy but also ensured a more accurate depiction of the research focus within the domain. Furthermore, keyword formatting was refined by capitalizing the first letter of each term to improve the visual quality and readability of the co-occurrence maps. Abbreviations such as ML (Machine Learning), RF (Random Forest), ANN (Artificial Neural Network), SVM (Support Vector Machine), and GIS (Geographic Information System) were retained to reflect common usage within the literature and enhance the interpretability of the network visualization.

Figure 2 shows a comprehensive overview of keyword relationships within the field of machine learning applications for groundwater potential studies. Using VOSviewer, the analysis identifies four distinct thematic clusters, each color-coded for clarity. The blue cluster highlights keywords associated with groundwater potential modeling, emphasizing approaches and terminology central to hydrological assessment. The green cluster focuses on GIS, reflecting the importance of spatial data handling and integration. The red cluster centers on machine learning algorithms, showcasing techniques such as random forests, support vector machines, and neural networks frequently employed in predictive modeling. Finally, the yellow cluster encompasses spatial prediction methods, bridging

geostatistical techniques with machine learning frameworks. Visual representation also encodes keyword frequency via node size and strength of association via edge thickness, allowing for intuitive interpretation of dominant themes and inter-keyword connectivity within the literature. This figure reveals the interdisciplinary nature of research in groundwater modeling, with clear convergence between geospatial analysis, data science, and environmental engineering.

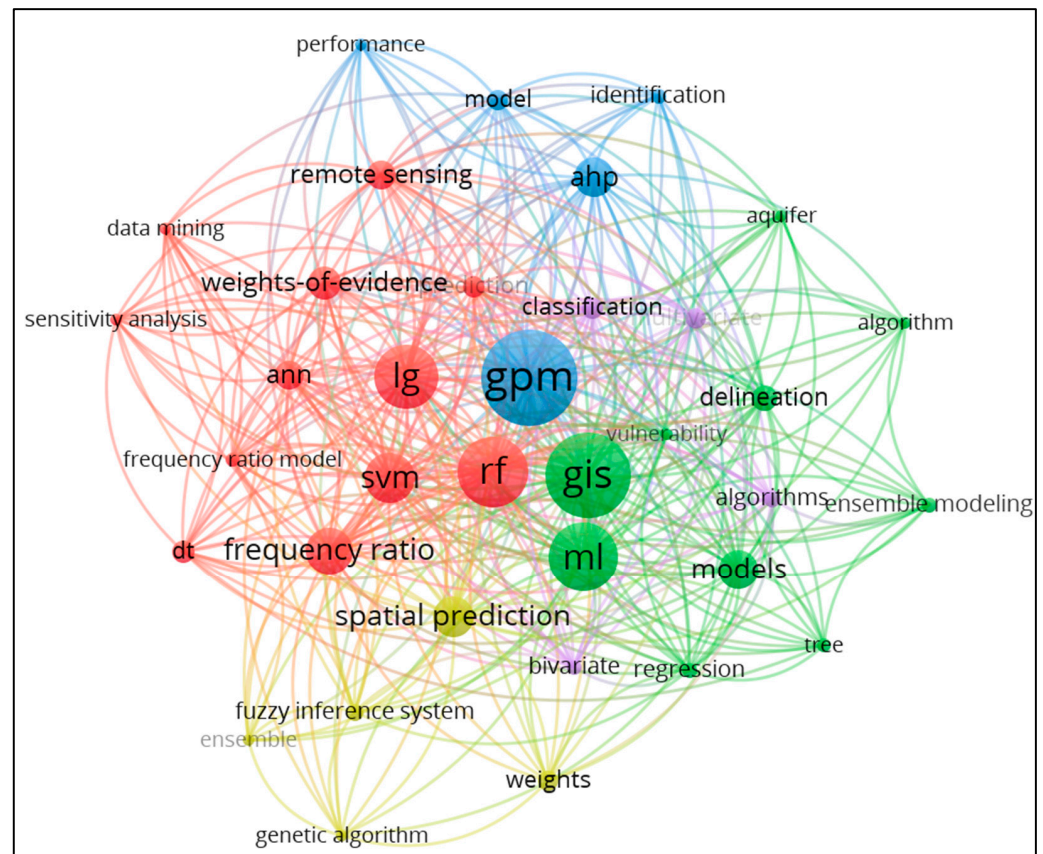


Figure 2. Network visualization map of keyword interactions related to machine learning applications in groundwater potential studies. Colored clusters represent distinct thematic groups derived from keyword co-occurrence analysis (blue: Groundwater potential mapping, green: geographic information systems, red: machine learning algorithms, yellow: spatial prediction methods). Node size corresponds to keyword frequency, while edge thickness reflects the strength of association between terms.

The overlay (time-stacked) visualization (Figure 3) shows a temporal progression in keyword usage. Earlier research, appearing in blue and green tones, focused primarily on GIS-based mapping and traditional statistical approaches. More recent publications, indicated in yellow, emphasize advanced ML techniques, including Deep Learning (DL) and hybrid models, signaling an evolving trend toward more sophisticated computational approaches. The increasing prominence of terms like feature selection, model validation, and ensemble learning suggests a growing emphasis on model performance and robustness in recent years.

These visualizations collectively demonstrate the dynamic and rapidly advancing landscape of groundwater potential modeling, where artificial intelligence is playing an increasingly pivotal role in supporting sustainable groundwater resource assessment and management.

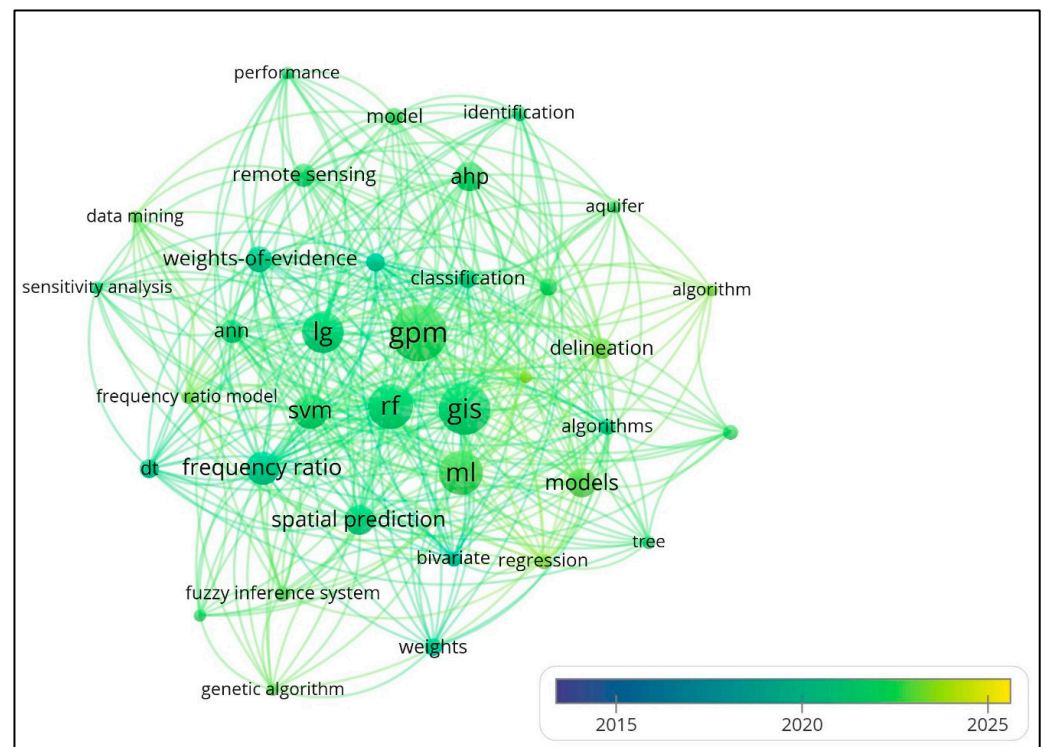


Figure 3. Temporal overlay visualization illustrating changes in keyword usage from 2015 to 2025, keyword nodes vary in size according to frequency, while the color gradient, from blue (earlier years) to yellow (recent years), depicts the chronological evolution of research themes.

3. Machine Learning Trends in GPM

3.1. Overview of Machine Learning in GPM

Machine Learning (ML) is a subset of artificial intelligence that enables computer systems to learn patterns and make data-driven predictions or decisions without being explicitly programmed for specific tasks [34,35]. In the context of GPM, ML algorithms have emerged as powerful tools for modeling complex and non-linear relationships between various physical, environmental, and anthropogenic factors influencing groundwater occurrence [36–38]. By leveraging large datasets and minimizing human bias, ML enhances the objectivity, reproducibility, and accuracy of GPM studies [39]. The process of applying ML in GPM typically follows a series of structured steps, as shown in Figure 4. These steps are covered in the next subsections in detail.

Over the past decade, a diverse array of ML algorithms has been employed in GPM, ranging from traditional decision tree-based models to advanced neural networks and hybrid optimization frameworks. The selection of these models is typically guided by the characteristics of the input data, the complexity of the hydrogeological terrain, and the desired balance between interpretability and predictive accuracy.

A comprehensive review of selected 83 studies reveals that ML models used in GPM can be systematically categorized into six functional groups (G1–G6), based on both algorithmic behavior and their applicability to hydrogeological analysis. Table 1 presents the classification of ML models commonly applied in GPM. Table 2 provides a cross-tabulation matrix showing the distribution of these model groups across the reviewed literature. Additionally, Figure 5 illustrates the frequency distribution of ML model groups, highlighting prevailing trends and preferences in recent GPM research.

Table 1. Machine learning (ML) models classification in groundwater potential mapping (GPM).

Group	ML Model	Acronym	Group	ML Model	Acronym
G1	Classification and Regression Tree	CART	G3	Deep Learning Neural Network	DLNN
	Credal Decision Tree	CDT		Deep Neural Network	DNN
	Decision Tree	DT		Long Short-Term Memory	LSTM
	Decision Tree Classifier	DTC		Multilayer Perceptron	MLP
	Deep Learning Tree	DLT		Neural Network	NN
	Decision Jungle	DJ	G4	Perceptron	PER
	Extra Trees Classifier	ETC		Locally Deep Support Vector Machine	LD-SVM
	Extremely Randomized Trees	ET		Nu-Support Vector Classifier	nuSVC
	Forest by Penalizing Attributes	FPA		Support Vector Classifier	SVC
	Logistic Model Tree	LMT		Support Vector Machine	SVM
	M5 Prime	M5P		Support Vector Machine for Regression	SMOReg
	Naïve Bayes Tree	NBT		Support Vector Regression	SVR
	Partial Decision Tree	PART		Bayesian Model Averaging	BMA
	Quick Unbiased Efficient Statistical Tree	QUEST	G5	Bayes Point Machine	BPM
	Random Forest	RF		Gaussian Process	GP
G2	Random Forest Classifier	RFC		Gaussian Process Classifier	GPC
	Random Tree	RT		Maximum Entropy	MaxEnt
	Self-Learning Random Forest	SLRF		Naïve Bayes	NB
	Split Point Sampling and Node Attribute Subsampling Classifier	SPAARC		Gaussian Naïve Bayes	NBA
	Averaged Perceptron	AP		K-Nearest Neighbor Classifier	KNC
	Dagging Logistic Regression	DLR		K-Nearest Neighbor	KNN
	Stochastic Gradient Descent	SGD		Linear Vector Quantization	LVQ
	Binary Logistic Regression	BLR		Linear Discriminant Analysis	LDA
	Logistic Regression	LG		Mixture Discriminant Analysis	MDA
	L1-Regularized Logistic Regression	LRL1	G6	Quadratic Discriminant Analysis	QDA
G3	Multivariate Adaptive Regression Splines	MARS		Rotation Forest Linear Discriminant Analysis	RFLDA
	Projection Pursuit Regression	PPR		Fuzzy Unordered Rule Induction Algorithm	FURIA
	Ridge Classifier	RID		Automated Machine Learning	AutoML
G3	Artificial Neural Network	ANN		Group Method of Data Handling	GMDH
	Artificial Neural Network Multi-Layer Perceptron	ANN-MLP		Passive-Aggressive Classifier	PASSI
	Automatic Multilayer Perceptron	AutoMLP		Zero Rule Classifier	ZeroR
	Convolutional Neural Network	CNN			

Notes: G1 = Tree/Ensemble, G2 = Regression, G3 = Neural Nets, G4 = Support Vector Machines, G5 = Bayesian, G6 = Other.

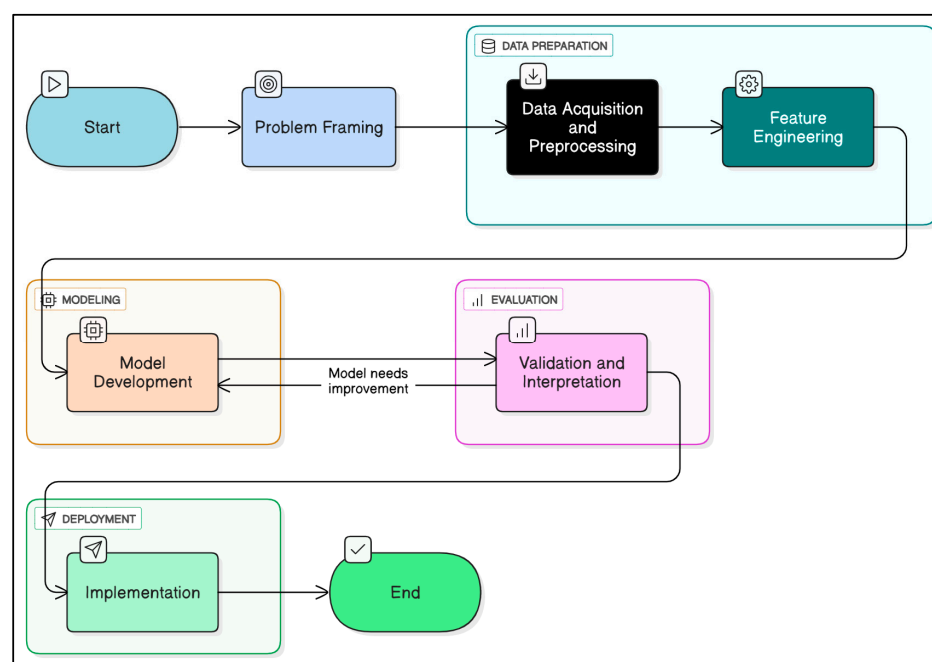


Figure 4. Machine learning workflow for groundwater potential mapping (GPM).

Table 2. Cross-tabulation matrix of machine learning model groups in the literature (83 papers).

Year	References	G1	G2	G3	G4	G5	G6	Year	References	G1	G2	G3	G4	G5	G6
2015	[40]	✓	✗	✗	✗	✗	✗	2022	[41]	✓	✓	✓	✓	✗	✗
2016	[42]	✓	✗	✗	✗	✗	✗	2022	[32]	✓	✗	✗	✗	✗	✗
2016	[43]	✓	✗	✗	✗	✗	✗	2022	[44]	✓	✗	✗	✗	✓	✗
2017	[45]	✗	✓	✗	✗	✗	✗	2022	[46]	✓	✓	✓	✓	✓	✓
2019	[47]	✓	✓	✗	✗	✗	✗	2022	[48]	✓	✓	✓	✓	✗	✗
2019	[49]	✓	✗	✓	✓	✗	✗	2022	[37]	✓	✗	✓	✗	✗	✓
2019	[50]	✓	✓	✗	✓	✗	✗	2023	[51]	✓	✗	✗	✗	✗	✗
2019	[52]	✓	✗	✗	✗	✗	✓	2023	[53]	✓	✓	✗	✗	✗	✗
2019	[54]	✗	✗	✗	✗	✗	✓	2023	[55]	✓	✗	✗	✗	✗	✗
2019	[56]	✓	✓	✗	✗	✗	✗	2023	[57]	✓	✗	✓	✗	✗	✓
2020	[58]	✓	✓	✗	✓	✗	✓	2023	[59]	✓	✗	✗	✗	✗	✗
2020	[60]	✓	✗	✗	✗	✗	✗	2023	[39]	✓	✗	✗	✓	✗	✗
2021	[61]	✓	✗	✗	✗	✗	✗	2023	[62]	✓	✗	✗	✗	✗	✗
2020	[63]	✓	✗	✗	✗	✗	✗	2023	[64]	✓	✗	✓	✓	✗	✗
2020	[65]	✗	✗	✓	✗	✗	✗	2023	[16]	✓	✗	✗	✗	✗	✓
2020	[66]	✓	✗	✗	✗	✗	✗	2024	[67]	✗	✓	✗	✗	✗	✗
2021	[68]	✓	✓	✗	✗	✗	✗	2020	[31]	✓	✗	✓	✗	✗	✗
2025	[69]	✓	✗	✗	✗	✗	✗	2024	[70]	✓	✗	✗	✗	✗	✗
2020	[66]	✓	✗	✗	✗	✗	✓	2024	[71]	✓	✓	✗	✓	✗	✗
2021	[72]	✗	✗	✓	✗	✗	✗	2023	[73]	✓	✗	✗	✗	✓	✓
2021	[14]	✓	✗	✗	✗	✗	✗	2023	[36]	✓	✗	✓	✓	✗	✗

Table 2. Cont.

Year	References	G1	G2	G3	G4	G5	G6	Year	References	G1	G2	G3	G4	G5	G6
2021	[61]	✔	✔	✔	✖	✖	✖	2023	[17]	✔	✖	✔	✔	✖	✔
2021	[74]	✔	✖	✔	✖	✖	✔	2023	[75]	✖	✖	✖	✔	✖	✖
2021	[76]	✖	✖	✖	✖	✖	✔	2024	[77]	✔	✖	✖	✖	✖	✖
2021	[78]	✖	✖	✖	✖	✔	✔	2024	[20]	✔	✔	✔	✖	✖	✖
2021	[79]	✔	✔	✖	✔	✖	✖	2024	[80]	✔	✖	✖	✔	✖	✖
2021	[81]	✔	✔	✔	✔	✔	✖	2024	[82]	✔	✖	✖	✔	✖	✔
2022	[83]	✔	✖	✔	✔	✖	✖	2024	[33]	✔	✖	✖	✖	✖	✖
2022	[32]	✔	✖	✖	✖	✖	✖	2024	[84]	✔	✖	✖	✖	✖	✖
2020	[85]	✔	✖	✖	✖	✖	✖	2024	[86]	✖	✖	✔	✖	✖	✖
2022	[87]	✔	✖	✔	✔	✖	✖	2024	[29]	✖	✖	✖	✔	✖	✖
2023	[88]	✔	✖	✔	✔	✔	✖	2025	[89]	✔	✖	✖	✖	✖	✖
2022	[90]	✔	✖	✖	✖	✖	✖	2025	[69]	✔	✔	✖	✖	✖	✔
2022	[91]	✔	✔	✖	✖	✖	✖	2025	[92]	✔	✖	✖	✔	✖	✖
2022	[93]	✔	✖	✔	✖	✖	✖	2025	[30]	✔	✖	✖	✖	✖	✖
2022	[94]	✖	✖	✖	✔	✖	✖	2025	[95]	✔	✖	✖	✔	✖	✖
2022	[21]	✔	✔	✖	✖	✖	✖	2025	[96]	✔	✖	✖	✔	✖	✖
2022	[97]	✔	✖	✖	✖	✔	✖	2025	[98]	✔	✖	✖	✖	✔	✔
2022	[99]	✖	✖	✖	✖	✖	✔	2025	[12]	✔	✖	✔	✔	✖	✖
2022	[100]	✔	✖	✖	✖	✖	✖	2025	[28]	✔	✖	✔	✔	✖	✖
2022	[101]	✔	✔	✔	✔	✖	✖	2025	[27]	✔	✖	✖	✖	✔	✖
2022	[18]	✔	✖	✖	✖	✖	✖								

Notes: ✔ refers to the model group used in this study, while ✖ refers to the model group not used.

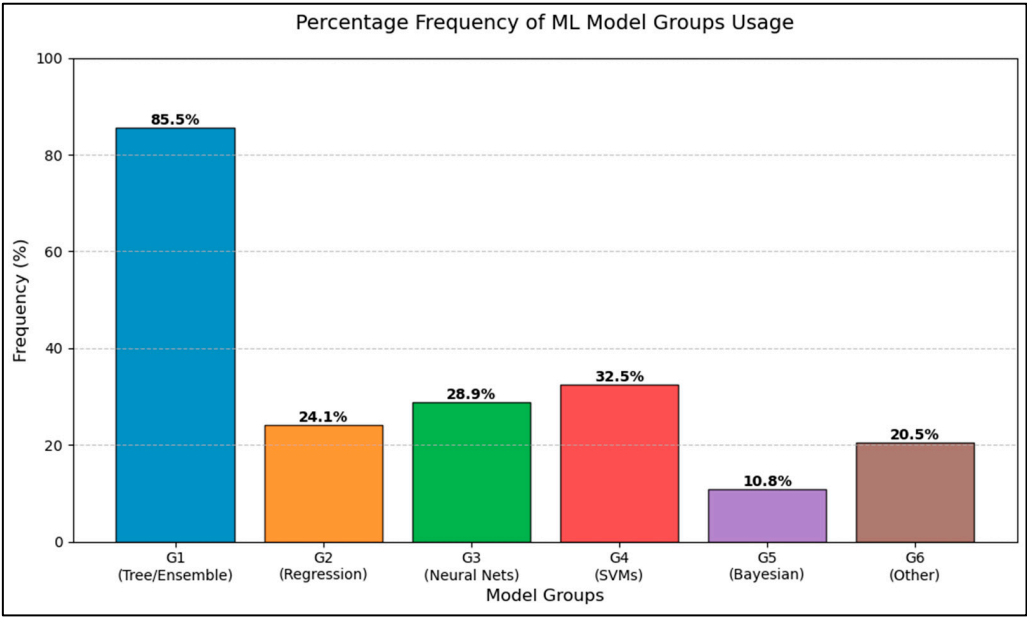


Figure 5. Frequency distribution of machine learning model groups used in groundwater potential mapping studies.

The cumulative evidence from Tables 1 and 2, and Figure 5 indicates strong methodological patterns in GPM studies. Tree-based and ensemble methods (G1), specifically Random Forest and Gradient Boosting models, appear to be the most popular ones, which is indicative of their ability to efficiently deal with nonlinear relationships, multicollinearity, and heterogeneous spatial data. Support Vector Machines (G4) and Neural Networks (G3) have also been used extensively, especially in research that aims for higher accuracy in GPM. Regression models (G2), specifically Logistic Regression, continue to be popular, often being used as a baseline model for comparison. The cross-tabulation analysis also reveals a rising trend towards the use of hybrid models, where two or more groups of models are integrated to improve predictive accuracy and uncertainty. The overall frequency distribution pattern, therefore, reveals a methodological shift from traditional statistical models towards more sophisticated ensemble and hybrid models in machine learning.

3.2. Problem Framing in ML-Based GPM

The application of ML in GPM requires a well-defined problem formulation to ensure alignment with hydrogeological objectives [102]. ML encompasses four primary paradigms: supervised, unsupervised, semi-supervised, and reinforcement learning [103]. In supervised learning, models are trained on labeled datasets, enabling them to learn from known input–output relationships [104]. In contrast, unsupervised learning explores unlabeled data to uncover inherent structures or patterns. Semi-supervised learning combines both labeled and unlabeled data to enhance learning performance, while reinforcement learning relies on a reward-based system to iteratively improve decision-making through interaction with the environment [90,105].

Among these paradigms, supervised learning is the dominant approach in ML-based GPM, as it aligns closely to predict groundwater occurrence or potential based on known examples (e.g., spring or well locations, or quantitative yield data). In supervised GPM applications, the predictive task can be framed as either a classification or regression problem, depending on the nature of the output variable. Classification involves predicting categorical groundwater potential zones (e.g., low, moderate, high). For example, ref. [40,106] trained Random Forest (RF) and Boosted Regression Tree (BRT) models using binary spring occurrence data (presence/absence), with the resulting probabilities classified into four groundwater potential zones. Regression, on the other hand, estimates continuous outcomes such as groundwater quality, level, yield, or recharge rates [107,108]. The choice between classification and regression largely depends on data availability. Classification models are preferable when only spatial occurrence data (e.g., spring or well locations) are available, while regression models require quantitative yield measurements. As noted by Hakim et al. [109], classification-based GPM remains more prevalent due to the wider availability of binary spring data; however, regression-based approaches are gaining traction with the increasing accessibility of detailed hydrogeological measurements.

3.3. Input Parameters and Feature Engineering

3.3.1. Types of Relevant Parameters Used in GPM

Various conditioning parameters have been widely used in ML-based GPM studies to represent the physical, geological, hydrological, and environmental factors influencing groundwater occurrence. Among these, geological, topographical, hydrological, and land use/land cover (LULC) variables are the most commonly applied. The selection of conditioning factors generally depends on the hydrogeological characteristics of the study area, the spatial resolution and availability of datasets, and the relevance of each parameter to groundwater recharge, storage, and movement. In the literature, these parameters have been classified using different frameworks. For instance, categorized

variables based on their source of information into five groups: cartography, satellite images, digital elevation models (DEMs), borehole data, and other sources. In contrast, ref. [110] proposed a classification based on regional characteristics, grouping variables into six categories: topographic, hydrological, forest type, soil, land use, and geological factors. Ref. [32] introduced a functional classification based on hydrogeological roles, including topographic, hydrological, hydrogeological, environmental, and climatic parameters. Based on groundwater recharge, the factors have been classified into five categories: climate, soil, LULC, and hydrology factors [17,46]. Similarly, Ref. [30] organized input variables into thematic layers such as lithology, geomorphology, soil, lineaments, LULC, and vegetation indices like NDVI. In this study, we adopt a comprehensive thematic classification scheme that includes nine categories: C1—Topographic, C2—Soil, C3—Hydrological, C4—Hydrogeological, C5—Hydrochemical, C6—Geological, C7—Environmental, C8—Climatic, and C9—Anthropogenic. Table 3 presents the most frequently used input parameters in GPM studies, organized by thematic group, while Table 4 provides a cross-tabulation matrix showing the distribution of these thematic categories across 83 reviewed studies. Additionally, Figure 6 illustrates the frequency distribution of these thematic categories in previous GPM studies, highlighting their relative importance in model development.

Table 3. Most frequently used input parameters in groundwater potential mapping (GPM) studies, categorized by thematic group.

Parameter	Category	Parameter	Category
Altitude	C1	Total hardness	C5
Slope		Sodium	
Valley depth		Potassium	
Terrain ruggedness index		Nitrate	
Curvature		Chloride	
Multi-resolution ridge top flatness (MRRTF)		Magnesium	
Multi-resolution valley bottom flatness (MRVBF)		Iron	C6
Roughness		Distance from faults	
Hillshade		Lineament density	
Soil type		Distance from lineaments	
Soil texture	C2	Lithology	C6
Clay content		Fracture density	
Silt		Basement depth	
Sand		Distance from paleochannels	
Soil depth		Fault density	
Soil hydrological group		Clay mineral alteration ratio	
Coarse fragments		Normalized Difference Vegetation Index (NDVI)	C7
Cation exchange capacity (CEC)		Land Use/Land Cover (LULC)	

Table 3. Cont.

Parameter	Category	Parameter	Category
Distance from rivers/water bodies	C3	Enhanced Vegetation Index (EVI)	C7
Drainage density		Normalized Difference Moisture Index (NDMI)	
River/Stream/Channel density		Built-up Index (BUI)	
Stream power index (SPI)		Bare Soil Index (BSI)	
Topographic Wetness Index (TWI)		Soil Adjusted Vegetation Index (SAVI)	
Flow accumulation		Vegetation type/cover	C8
Spring density		Rainfall/Precipitation	
Runoff depth		Temperature	
Distance from stream/channel		Evapotranspiration	
Depth to groundwater table		Relative Humidity	
Aquifer type	C4	Snow depth	C9
Hydraulic head		Climate type	
Transmissivity		Distance from roads	
Specific yield		Distance to villages/residential areas	
Well depth		Population density	
Groundwater table elevation		Cropping intensity	
Total water storage		Irrigated area	C9
Electrical conductivity (EC)		Distance to qanats	
pH			

Notes: C1 = Topographic, C2 = Soil, C3 = Hydrological, C4 = Hydrogeological, C5 = Hydrochemical, C6 = Geological, C7 = Environmental, C8 = Climatic, C9 = Anthropogenic.

Table 4. Cross-tabulation matrix of data thematic groups in literature (83 papers).

Year	References	C1	C2	C3	C4	C5	C6	C7	C8	C9	Year	References	C1	C2	C3	C4	C5	C6	C7	C8	C9
2015	[40]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2022	[41]	✓	✓	✓	✗	✓	✓	✓	✓	✓
2016	[42]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2022	[32]	✓	✓	✓	✗	✓	✗	✓	✓	✓
2016	[43]	✓	✓	✓	✓	✓	✓	✗	✗	✗	2022	[44]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2017	[45]	✓	✓	✗	✗	✓	✗	✗	✗	✗	2022	[46]	✓	✓	✓	✓	✓	✓	✓	✓	✗
2019	[47]	✓	✓	✓	✗	✓	✓	✓	✓	✓	2022	[48]	✓	✓	✓	✗	✓	✗	✓	✓	✗
2019	[49]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2022	[37]	✓	✓	✓	✗	✓	✓	✓	✓	✓
2019	[50]	✓	✓	✓	✗	✓	✓	✓	✓	✓	2023	[51]	✓	✓	✗	✗	✗	✗	✓	✓	✗
2019	[52]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2023	[53]	✓	✓	✓	✗	✗	✓	✓	✓	✗
2019	[54]	✓	✓	✓	✗	✓	✓	✓	✗	✓	2023	[55]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2019	[56]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2023	[57]	✓	✓	✓	✓	✓	✓	✓	✓	✗
2020	[58]	✓	✓	✓	✗	✓	✓	✗	✓	✗	2023	[59]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2020	[60]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2023	[39]	✓	✓	✓	✗	✓	✓	✓	✓	✓
2021	[61]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2023	[62]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2020	[63]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2023	[64]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2020	[65]	✓	✓	✓	✗	✓	✗	✓	✓	✗	2023	[16]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2020	[66]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2024	[67]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2021	[68]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2020	[31]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2025	[69]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2024	[70]	✓	✓	✓	✗	✓	✓	✓	✓	✓

Table 4. Cont.

Year	References	C1	C2	C3	C4	C5	C6	C7	C8	C9	Year	References	C1	C2	C3	C4	C5	C6	C7	C8	C9
2020	[66]	✓	✓	✗	✓	✓	✓	✗	✗	✗	2024	[71]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2021	[72]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2023	[73]	✓	✓	✓	✗	✓	✓	✗	✓	✗
2021	[14]	✓	✓	✓	✓	✓	✓	✗	✗	✗	2023	[36]	✓	✓	✓	✓	✓	✓	✓	✓	✗
2021	[61]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2023	[17]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2021	[74]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2023	[75]	✓	✓	✓	✗	✓	✓	✗	✓	✗
2021	[76]	✓	✓	✓	✗	✓	✗	✓	✓	✗	2024	[77]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2021	[78]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2024	[20]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2021	[79]	✓	✓	✓	✓	✓	✓	✓	✓	✗	2024	[80]	✓	✓	✓	✗	✗	✓	✓	✓	✗
2021	[81]	✓	✓	✓	✓	✓	✓	✓	✗	✗	2024	[82]	✓	✓	✓	✗	✓	✗	✓	✓	✗
2022	[83]	✓	✓	✗	✗	✗	✗	✓	✓	✗	2024	[33]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2022	[32]	✓	✓	✓	✗	✓	✗	✓	✓	✓	2024	[84]	✓	✓	✓	✓	✓	✓	✓	✓	✓
2020	[85]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2024	[86]	✓	✓	✓	✗	✓	✗	✓	✓	✗
2022	[87]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2024	[29]	✓	✓	✓	✗	✓	✓	✓	✗	✗
2023	[88]	✓	✓	✓	✗	✓	✗	✗	✓	✗	2025	[89]	✓	✓	✓	✓	✗	✗	✓	✓	✗
2022	[90]	✓	✓	✓	✓	✓	✓	✓	✓	✗	2025	[69]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2022	[91]	✓	✓	✓	✓	✓	✓	✓	✓	✗	2025	[92]	✓	✓	✓	✗	✓	✓	✓	✓	✓
2022	[93]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2025	[30]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2022	[94]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2025	[95]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2022	[21]	✓	✓	✓	✗	✓	✓	✓	✓	✓	2025	[96]	✓	✓	✓	✓	✓	✓	✓	✓	✗
2022	[97]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2025	[98]	✓	✓	✓	✗	✓	✗	✓	✓	✗
2022	[99]	✓	✓	✓	✗	✓	✓	✓	✓	✗	2025	[12]	✓	✓	✗	✗	✗	✗	✓	✓	✗
2022	[100]	✓	✓	✓	✗	✓	✓	✓	✗	✗	2025	[28]	✓	✓	✓	✓	✓	✓	✓	✓	✗
2022	[101]	✓	✓	✓	✗	✓	✓	✓	✓	✓	2025	[27]	✓	✓	✓	✗	✓	✓	✓	✓	✗
2022	[18]	✓	✓	✓	✗	✓	✗	✓	✓	✗											

Notes: C1 = Topographic, C2 = Soil, C3 = Hydrological, C4 = Hydrogeological, C5 = Hydrochemical, C6 = Geological, C7 = Environmental, C8 = Climatic, C9 = Anthropogenic. ✓ = Data thematic group used in this study; ✗ = Data thematic group not used.

As illustrated in Figure 6, the proportion of parameters classified under C4—Hydrogeological appears relatively low compared to other thematic groups. This underrepresentation is largely attributed to the common practice in the literature of merging hydrogeological parameters, such as aquifer type, transmissivity, and groundwater table depth, with hydrological factors like drainage density and flow accumulation into a single “Hydrological/Hydrogeological” category [111–113]. Such merging often occurs when datasets originate from mixed sources, including surface water data derived from RS and subsurface information obtained from borehole records or pumping tests. In many studies, the distinction between surface hydrological processes and subsurface groundwater properties is not explicitly made, leading to a blurred classification and reduced visibility of hydrogeological parameters as a standalone category. This trend is further reinforced by the methodological challenges associated with acquiring detailed hydrogeological data. Unlike hydrological parameters, which can be extracted from widely available topographic and satellite datasets, hydrogeological variables often require field-based measurements, geophysical surveys, or long-term monitoring, making them less accessible and more resource-intensive to obtain [30,114]. Additionally, hydrogeological mapping demands a high level of conceptual integration and cartographic reasoning, which is not always feasible in data-driven ML-based GPM studies that prioritize computational efficiency over geological depth [114,115].

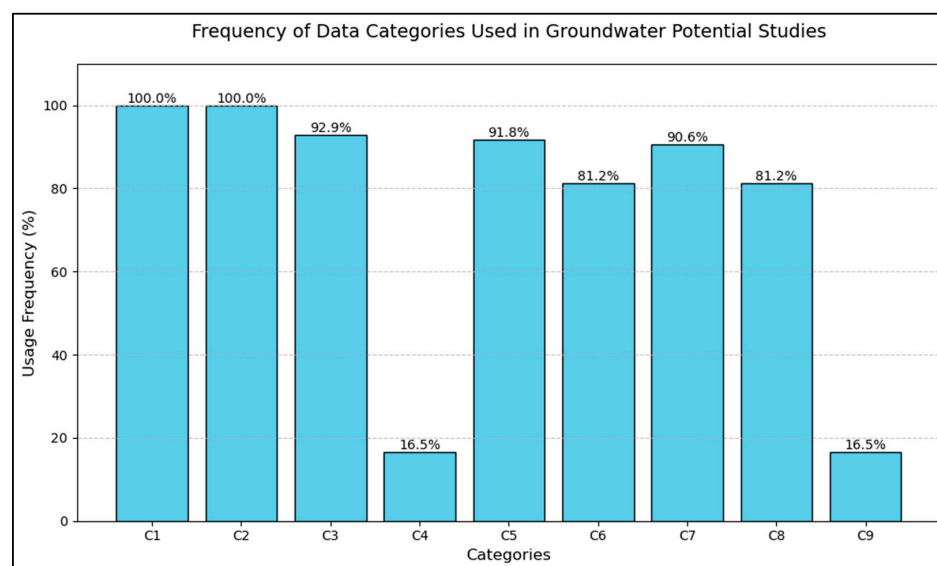


Figure 6. Frequency of data categories used in groundwater potential mapping studies. C1 = Topographic, C2 = Soil, C3 = Hydrological, C4 = Hydrogeological, C5 = Hydrochemical, C6 = Geological, C7 = Environmental, C8 = Climatic, C9 = Anthropogenic.

In GPM using RS and GIS techniques, distance from river and proximity to water bodies like lakes and water bodies are two of the most important hydrological conditioning factors; however, there is a slight variation in groundwater recharge mechanisms in these two parameters. It is well understood that distance from river is one of the most important parameters in groundwater potential mapping because river systems support surface water accumulation and infiltration into groundwater [87]. Therefore, groundwater potential is inversely proportional to distance from river. Groundwater potential is high in areas within 0–500 m of river channels [116]. On the other hand, water bodies like lakes, ponds, and wetlands support groundwater levels through surface water and groundwater interaction; however, it is a complicated process since water bodies may behave like discharge areas if the water table is above the lake level [117]. From a spatial modeling viewpoint, distance from river is represented as a Euclidean distance surface for groundwater potential mapping; however, proximity to water bodies is represented as buffer distances or land use/land cover classifications. Although river proximity is given more importance in groundwater potential mapping due to dynamic recharge potential, proximity to water bodies like lakes and water bodies may indicate areas of high groundwater potential. In reality, areas of high groundwater potential are often associated with favorable conditions like low slope values, high lineament density values, and proximity to river and water bodies [28].

The reliability of GPM models also depends on the quality, type, and spatial representativeness of the input data used for model development. Generally speaking, GPM studies are based on two primary data components: (1) groundwater inventory data and (2) conditioning factors. Groundwater inventory data may include high-yield well point data [23,30,91,118], borehole productivity data [80,86,90], spring occurrences [56,119], groundwater level measurements [44,96], and sometimes transmissivity or specific capacity values [120]. The number of inventory data points plays a crucial role in the development of the GPM, as it significantly affects the model's accuracy, robustness, and predictive capability. A larger dataset improves the representation of spatial variability and hydrogeological conditions, reduces uncertainty, and minimizes the risk of overfitting. Consequently, adequate inventory data are essential for ensuring reliable calibration and validation of the model (data density calculated from previous GPM studies is listed in Table A1).

Although the number of inventory data points can affect model stability, representativeness may play a more crucial role; several studies suggest that the distribution of inventory data samples over drainage basins or hydrogeological units (number of pixels per watershed) may be more important for reliable prediction model development than the sample size itself [87,91,96].

Apart from the above data set, the second set of data used for model development includes conditioning factors such as those extracted from geological maps, DEM products, RS products, and climatic data sets [56,80,86,90,119]. Some of the commonly used conditioning factors include lithology, slope, elevation, drainage density, lineament density, rainfall, land use/land cover, topographic wetness index, etc. [23,30,56,91,119,120]. Moreover, proximity to surface water bodies such as rivers, lakes, reservoirs, or large water surfaces represents an important hydrological conditioning factor used in GPM model development, as these areas are potential recharge zones for the water table [28,87,116,117].

3.3.2. Data Sources and Preprocessing

Accurate and reliable data sources are fundamental to the development of robust GPMs using ML. Each conditioning parameter is typically derived from specific datasets, often obtained through a combination of RS products, thematic maps, and in situ measurements. Table 5 summarizes the key data sources most frequently employed in GPM studies. Topographic and terrain-related parameters, such as slope or the TWI, are commonly extracted from DEMs provided by SRTM, ASTER, or ALOS [31,37,76,81,100]. Land cover and vegetation indices are derived from multispectral satellite imagery such as Landsat, Sentinel-2, or MODIS [12,83,94,121]. Geological and soil information is sourced from national geological surveys and global repositories such as FAO, ISRIC, and OneGeology [32,88]. Hydrological network data, including river and stream layers, are often obtained from GIS-based datasets like HydroSHEDS. Hydrogeological parameters, including borehole and aquifer characteristics, are typically collected from groundwater authorities or field surveys. Climate-related inputs, such as precipitation and temperature, are sourced from global climate databases such as CHIRPS, WorldClim, and TRMM. By integrating these diverse datasets, GPM studies ensure that both spatial and thematic variability are adequately captured, thereby enhancing the predictive capacity and generalizability of ML models.

Table 5. Key data sources are commonly used in GPM.

Parameter Type	Common Data Sources	Links
DEM-derived (e.g., slope, TWI)	SRTM, ASTER, ALOS DEM	https://earthexplorer.usgs.gov/ (accessed on 29 January 2025) https://asterweb.jpl.nasa.gov/gdem.asp (accessed on 21 December 2025) https://www.eorc.jaxa.jp/ALOS/en/index_e.htm (accessed on 20 December 2025)
Land cover & vegetation	Landsat, Sentinel-2, MODIS	https://www.usgs.gov/landsat-missions (accessed on 20 December 2025) https://modis.gsfc.nasa.gov/data/ (accessed on 19 December 2025)
Geological & soil maps	National geological surveys, FAO, ISRIC	https://aims.fao.org/news/fao-geonetwork-free-access-spatial-datasets-and-related-interactive-digital-maps (accessed on 20 December 2025) https://soilgrids.org/ (accessed on 19 December 2025) https://onegeology.org/ (accessed on 19 December 2025)
Hydrological networks	GIS-based river layers, HydroSHEDS	https://www.hydrosheds.org/ (accessed on 20 December 2025)
Borehole & aquifer data	Groundwater authorities, field surveys	https://www.bgs.ac.uk/ (accessed on 20 December 2025)
Climate data	CHIRPS, WorldClim, TRMM	https://www.chc.ucsb.edu/data/chirps (accessed on 21 December 2025) https://www.worldclim.org/ (accessed on 22 December 2025) https://disc.gsfc.nasa.gov/datasets?keywords=TRMM (accessed on 22 December 2025)

The analytical stage of GPM involves integrating spatial processing techniques with GIS, along with implementing machine learning models. In the majority of the research articles, thematic layers of conditioning data were prepared using GIS platforms like ArcGIS, QGIS, etc. In the majority of the research articles, the modeling of the machine learning model was performed using Python or R environments with the help of open-source libraries like scikit-learn, XGBoost, TensorFlow, caret, etc. In a few research articles, proprietary statistical software packages were utilized. In spite of the detailed discussion of the methodological approaches, the sharing of the source code or scripts of the research articles is very limited. In recent research articles, a gradual shift towards the utilization of open-source platforms is evident. This will definitely enhance the overall methodological rigor of the research articles.

3.3.3. Feature Engineering

The effectiveness of ML in GPM critically depends on rigorous parameter selection and feature engineering, as these processes collectively determine both model accuracy and interpretability. Well-designed feature engineering not only enhances predictive performance but also reduces the complexity of the prediction models by minimizing redundancy and noise in the dataset [17,122]. This section systematically reviews techniques commonly employed in ML-based GPM studies, while highlighting essential considerations for variable selection, spatial data preprocessing, and feature optimization. In most applications, feature engineering can be divided into three primary steps: transforming raw spatial data, selecting relevant features, and standardizing inputs.

Transforming Raw Spatial Data

Raw spatial data obtained from diverse sources (e.g., DEMs, satellite imagery, borehole records, and climatic databases) often require transformation before they can be used as input features in ML models. DEMs are typically processed to generate secondary topographic derivatives such as slope, aspect, curvature, TWI, and SPI, which serve as proxies for terrain-driven hydrological processes. Satellite images (e.g., Landsat, Sentinel-2, MODIS) are commonly transformed into spectral indices such as the NDVI, Soil-Adjusted Vegetation Index (SAVI), and NDMI, which represent vegetation vigor, soil moisture, and land cover conditions. Geological and soil maps are digitized, reclassified, and converted into categorical layers to represent lithology, geomorphology, or soil type. Hydrological layers such as river networks, drainage density, flow accumulation, and spring density are generated through GIS-based spatial hydrological analysis.

To enhance their suitability for ML modeling, these raw datasets are frequently transformed into quantitative variables that explicitly capture their relationship with groundwater occurrence. Statistical techniques such as Frequency Ratio (FR) and Normalized Frequency Ratio (NFR) are employed to assess the degree of association between parameter classes and groundwater evidence locations (e.g., wells, springs) [121,123]. Similarly, probabilistic and information-based approaches, including the Weight-of-Evidence (WoE) [11,124], Information Value (IV) [125], and Evidence Belief Function (EBF) [87] methods, have been widely applied to quantify the predictive power of conditioning factors. These approaches assign numerical weights or probability values to thematic classes (e.g., lithological units, soil categories, land cover types), thereby transforming categorical data into interpretable continuous predictors. By systematically applying these transformations, raw data are converted into standardized, quantitative variables that capture their hydrogeological significance. This step ensures that the input features better represent the spatial and statistical relationships governing groundwater occurrence, thereby improving both the robustness and interpretability of ML- GPM.

Selecting Relevant Features

Not all extracted variables contribute equally to the predictive accuracy of ground-water potential models. Feature selection is therefore essential to eliminate redundant or irrelevant parameters that may increase model complexity and reduce generalizability. In GPM studies, statistical correlation analysis (e.g., correlation matrix (CM) results, Pearson's r , variance inflation factor (VIF), tolerances (Tol), Entropy method, and mutual information (MI) test) is often employed to detect multicollinearity among predictors [17,126,127]. Advanced feature selection methods include information gain, Relief-F, recursive feature elimination (RFE), and wrapper-based techniques integrated with ML algorithms such as RF or SVM [128–130]. Recent studies have also employed dimensionality reduction approaches such as Principal Component Analysis (PCA) to condense high-dimensional datasets into fewer uncorrelated components [70,131–134]. The choice of feature selection method depends on the nature of the dataset, the type of ML algorithm, and the desired balance between model interpretability and accuracy. Selecting the most relevant parameters not only improves computational efficiency but also ensures that the derived GPM reflects hydrogeological reality rather than artifacts of overfitting. Table 6 shows a summary of these common feature selection methods and their thresholds and decision rules in ML-based GPM studies.

Table 6. Common feature selection methods and their thresholds and decision rules in machine learning-based ground potential mapping (GPM) studies.

Method	Indicator/Metric	Threshold/Range	Decision Rule	Application in GPM	Advantages	Limitations	References
CM	Pearson's r	R	≥ 0.7 –0.9 Remove	Detects pairwise correlation among parameters	Simple, fast, interpretable	Only captures linear relationships	[17]
VIF	VIF score	1 = no collinearity; 1–5 = moderate; ≥ 10 = high	Remove variables with VIF ≥ 10	Identifies multicollinearity among predictors	Robust for multicollinearity analysis	Threshold choice is subjective	[17]
Tol	1/VIF	Tol ≤ 0.1	Indicates severe multicollinearity $\rightarrow$ remove	Identifies multicollinearity among predictors	Robust for multicollinearity analysis	Threshold choice is subjective	[17]
MI	MI score	MI > 0 = relevant; MI ≤ 0 = irrelevant	Retain positive MI; eliminate negative or near-zero	Measures dependence between input and target	Captures nonlinear relationships	Computationally expensive for large data	[135]
EW	Entropy value	High entropy = less useful; Low entropy = informative	Eliminate factors with an entropy weight ≈ 0	Evaluates the contribution of each factor to GPM	Retains the physical meaning of variables	Sensitive to noisy inputs	[127]
RFE/R-F	Ranked importance	No fixed cutoff; empirical	Retain top-ranked features (e.g., 5–10 most relevant)	Estimates the importance of attributes by instance weights	Works well with noisy data	May overlook weak predictors	[129]
PCA	Eigenvalue & variance	Eigenvalue ≥ 1 (Kaiser's rule); Cumulative variance ≥ 70 –90%	Retaining principal components meeting criteria	Reduces feature space into uncorrelated components	Handles high-dimensional datasets	Loss of interpretability, hydrogeological meaning	[131]

Notes: CM = Correlation Matrix, VIF = Variance Inflation Factor, Tol = Tolerance, MI = Mutual Information, EW = Entropy Weighting, RFE = Recursive Feature Elimination, R-F = Relief-F, PCA = Principal Component Analysis.

Apart from the statistical and wrapper methods of feature selection, the use of Random Forest (RF), an ensemble-based model, offers a direct way of assessing the importance of

predictors using out-of-bag (OOB) importance analysis. The OOB method relies on the bootstrap sampling technique, where about one-third of the data is left out of each tree during the construction process and is used to assess the performance of the model. The importance of the predictors is usually measured using the mean decrease in accuracy or mean decrease in impurity. The advantage of using OOB importance is that it is calculated during the training of the model without the need for an external validation data set, making it a computationally efficient and statistically sound approach for the identification of important groundwater potential conditioning factors. This approach has been used in GPM to rank variables and remove less informative predictors, making the model more interpretable and less prone to over-fitting.

Standardizing Inputs

The final stage of feature engineering involves standardizing or normalizing input variables to ensure that all features contribute proportionally during model training. Since GPM datasets often combine heterogeneous parameters (e.g., elevation in meters, rainfall in millimeters, and NDVI as unitless indices), differences in scale can bias algorithms that are sensitive to magnitude variations, such as k-Nearest Neighbor (KNN), SVM, or neural networks. Common normalization techniques include min–max scaling (rescaling features to a 0–1 range), z-score standardization (mean-centered scaling with unit variance), and log transformations (for positively skewed variables such as drainage density) [46,84]. Standardization not only accelerates convergence during training but also improves numerical stability and reduces the dominance of high-magnitude predictors. Some ensemble methods (e.g., Random Forest, Gradient Boosting) are less sensitive to scaling; however, applying consistent standardization across the dataset is considered best practice in ML-based GPM.

3.4. Model Development and Validation Techniques

The development of ML models for GPM involves several critical steps, including model selection, training, hyperparameter optimization, and performance validation. These stages collectively determine the model's predictive accuracy, generalizability, and reliability in identifying groundwater potential zones. The overarching goal is to ensure that the final model not only exhibits strong statistical performance but also aligns with the underlying hydrogeological and environmental processes. Additionally, it must also perform well for independent datasets. Table A2 provides a comprehensive summary of representative studies that demonstrate the methodological aspects of ML model development and validation in GPM conducted between 2015 and 2025. The table highlights key elements of each study, including dataset characteristics, dataset division strategy, tuning and learning methods, evaluation metrics, and model explainability techniques.

3.4.1. Model Selection and Training

The selection and training of ML models constitute the foundation of GPM and significantly influence the predictive reliability and transferability of results. Based on the reviewed literature (Table A2), most GPM studies employed supervised learning algorithms, as the training datasets typically consist of known groundwater presence (e.g., springs, wells) and absence or non-yielding locations. These datasets enable algorithms to learn the complex relationships between hydro-environmental factors and groundwater occurrence. A wide spectrum of ML models has been applied to GPM as discussed in Section 3 and listed in Table 1, including tree-based ensemble methods, neural network architectures, support vector machines (SVMs), and probabilistic or statistical learners. Among these, RF and Boosted models (e.g., AdaBoost, Gradient Boosting, and XGBoost) are the most frequently adopted (Figure 5) due to their robustness to overfitting, capacity to handle non-linear interactions, and ability to rank feature importance [12,42,49]. LR

and SVM remain common benchmark models, offering transparency and interpretability. Recent studies have expanded toward deep learning frameworks such as CNNs and DNNs for capturing complex spatial dependencies and non-linearities in heterogeneous terrain [31]. In most reviewed studies, data were divided into training and testing subsets, typically using a 70:30 or 75:25 ratio, to train the model and validate its predictive capacity (e.g., [21,40,45]). Some studies adopted more rigorous approaches, such as k-fold cross-validation or stratified sampling, which repeatedly partition the data into multiple training and testing folds to obtain stable performance metrics. For instance, refs. [49,94] employed multi-fold validation to reduce sampling bias in datasets with uneven spatial distributions of groundwater points.

3.4.2. Hyperparameter Optimization

Hyperparameter optimization plays a pivotal role in enhancing the predictive performance and stability of ML models for GPM. Hyperparameters, such as the number of estimators in RF, learning rate in GB, kernel type and regularization parameter in SVM, or the number of neurons and layers in ANNs, determine how effectively a model learns from data. Proper tuning of these parameters reduces overfitting, enhances generalization to unseen regions, and improves computational efficiency. Across GPM literature, optimization techniques have evolved significantly. Earlier studies [40,47] often relied on manual or trial-and-error tuning, adjusting parameters iteratively based on validation accuracy or area under the receiver operating characteristic curve (AUC). Although simple, this approach is subjective and computationally inefficient, especially when dealing with large feature sets and complex models. Subsequent studies adopted more systematic strategies, such as Grid Search and Random Search, which automate the tuning process by testing multiple parameter combinations within predefined ranges. Grid Search offers exhaustive exploration but can be computationally intensive, while Random Search provides a faster, probabilistic alternative with comparable accuracy. Examples include [21,32,101], who used these methods to optimize parameters for RF, GB, and AutoMLP models in groundwater prediction tasks.

Recent developments demonstrate a clear shift toward metaheuristic and Bayesian Optimization (BO) frameworks. These adaptive algorithms iteratively search for the hyperparameter space using prior performance knowledge, reducing computational cost while maintaining accuracy. Techniques such as BO, Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Randomized Search, and Teaching–Learning-Based Optimization (TLBO) have been applied to fine-tune model parameters in ensemble and hybrid frameworks [66,70,75]. Bayesian optimization, in particular, has shown superior performance in tuning complex nonlinear models such as CatBoost, XGBoost, and deep neural networks, achieving efficient convergence with fewer iterations. The increasing integration of AutoML pipelines has further transformed hyperparameter optimization in GPM research. AutoML frameworks, such as those applied by [37,70,107], automate not only hyperparameter tuning but also model selection, feature preprocessing, and validation, providing a reproducible and objective workflow that minimizes human bias. Despite these advances, challenges persist. Optimization results can be sensitive to data partitioning, regional variability, and the spatial heterogeneity of groundwater systems. Therefore, researchers increasingly employ cross-validation-based tuning (e.g., k-fold or nested cross-validation) to ensure robustness and prevent overfitting to local anomalies. The use of multi-objective optimization, balancing accuracy, computational cost, and interpretability, represents an emerging frontier in GPM modeling. In summary, the progression from manual adjustment to intelligent, automated, and metaheuristic optimization methods marks a crucial evolution in ML-based GPM. These approaches have improved model robustness,

reduced computational burden, and enhanced the transparency of decision-making in spatial groundwater assessments.

3.4.3. Model Validation and Performance Assessment

Model validation is an essential phase to assess the generalizability and predictive capacity of ML-based GPM models. It involves partitioning the dataset into training and testing subsets or applying k-fold cross-validation (typically 5-fold or 10-fold) to reduce bias due to random data splits, as discussed in the previous section. The performance of classification-based GPM models is commonly evaluated using both threshold-independent and threshold-dependent metrics. Threshold-independent metrics such as the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) provide an overall measure of the model's discriminative ability. Threshold-dependent metrics include Accuracy, Precision, Recall (Sensitivity), Specificity, and F1-score, which evaluate the balance between correctly and incorrectly classified instances. For regression-based GPM models, performance is typically assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Squared Error (MSE), and the Coefficient of Determination (R^2). The ROC and its corresponding AUC were the most widely adopted indicators, appearing in over 80% of studies (e.g., [42,61,95]). An AUC > 0.9 generally indicates excellent discrimination, while values between 0.7–0.9 reflect good to moderate model performance. Complementary metrics such as accuracy, precision, recall, specificity, and the F1-score were used to capture both correct detections and false predictions of groundwater occurrence [16,101]. Error-based measures, including RMSE, MAE, and MSE, were applied in regression-based models to evaluate continuous groundwater yield or recharge predictions (e.g., [64,100]). Some studies additionally incorporated Kappa statistics, Matthews correlation coefficient (MCC), and Nash–Sutcliffe efficiency (NSE) to assess the consistency between predicted and observed groundwater zones. Comparative validation techniques such as Success Rate Curves (SRCs), Prediction Rate Curves (PRCs), and Seed Cell Area Index (SCAI) have been used to evaluate the spatial agreement between predicted groundwater potential zones and known well or spring locations [49,69]. SCAI is a quantitative metric widely used to evaluate the spatial consistency and reliability of susceptibility and potential mapping results [47,49,99]. The concept was originally introduced by [136] to assess the relationship between the spatial distribution of observed events and the predicted susceptibility classes. In GPM studies, SCAI measures the ratio between the percentage of the total area occupied by a given groundwater potential class and the percentage of groundwater inventory points (e.g., wells or springs) located within that class [47,49,69,99,137]. A reliable model is expected to produce lower SCAI values for high and very high potential zones, indicating that most groundwater occurrences are concentrated within relatively small areas predicted as favorable for groundwater accumulation [49,69]. Conversely, higher SCAI values are typically associated with low-potential zones, reflecting fewer groundwater occurrences relative to their spatial extent [49,69]. Therefore, a systematic decrease in SCAI values from low to high groundwater potential classes indicates a logical and consistent spatial prediction pattern. Several studies have applied this metric to evaluate groundwater potential maps produced using machine learning models and GIS-based approaches (e.g., [47,49,69,99,137]).

However, SCAI is sensitive to the number and spatial distribution of inventory data and may be unstable when sample sizes are limited or unevenly distributed. In comparison, threshold-independent metrics such as ROC–AUC assess overall discriminatory power but do not explicitly account for spatial allocation efficiency [137]. Accuracy, precision, recall, and F1-score provide classification-based performance measures but depend on threshold selection. Consequently, combining statistical metrics (e.g., AUC) with spa-

tial metrics (e.g., SCAI) is recommended to ensure comprehensive validation of GPM outputs [40,42,43,47,66]. Additionally, statistical significance tests, such as the Wilcoxon signed-rank test and Friedman test, were utilized by [32,41] to determine whether observed performance differences among multiple ML algorithms were statistically meaningful.

Although the ROC curve is widely used to evaluate the predictive performance of GPM models, it has certain limitations when used alone [42,43]. ROC analysis mainly assesses the ability of a model to discriminate between groundwater presence and absence based on the relationship between true positive and false positive rates. However, this metric does not explicitly evaluate the spatial consistency or distribution pattern of predicted groundwater potential classes across the study area [45,52]. As a result, models with similar ROC values may still produce different spatial prediction patterns. To overcome this limitation, ROC analysis is often complemented by the SCAI, which measures the relationship between the spatial extent of groundwater potential classes and the distribution of groundwater inventory points [45,52]. While ROC evaluates the overall predictive accuracy, SCAI provides additional insight into the spatial reliability and logical distribution of predicted groundwater potential zones. Therefore, the combined use of ROC and SCAI allows for a more robust and comprehensive interpretation of GPM results [40,42,43,47,61,66,78].

Beyond statistical validation, spatial validation and field verification are critical for evaluating the real-world applicability of the model. This involves comparing predicted groundwater potential zones with independent datasets such as newly identified wells, spring discharge locations, or hydrogeological survey results not used during model training. Spatial overlay analysis in GIS allows for the quantification of prediction accuracy across zones, providing insight into the spatial consistency and practical relevance of the GPM. Recent works have emphasized the integration of spatial cross-validation (SCV), which accounts for spatial autocorrelation among training and test samples, thereby reducing overoptimistic accuracy estimates often found in random cross-validation.

3.4.4. Model Interpretation and Explainability

Interpreting the internal mechanisms of ML models and understanding how each input variable influences the groundwater potential outcome are crucial steps in ensuring that the resulting maps are both scientifically valid and practically useful. Model interpretation and explainability not only enhance transparency but also help bridge the gap between data-driven predictions and hydrogeological reasoning, transforming black-box algorithms into interpretable tools for decision-making. Feature importance analysis is the most widely adopted approach for interpreting ML-based GPM models. It quantifies the relative influence of each conditioning factor on model prediction, thereby identifying the most critical drivers of groundwater occurrence. Traditional methods, such as permutation feature importance (PFI) and mean decrease in impurity (MDI), are embedded in tree-based algorithms like RF, GBM, and XGB. These techniques evaluate how the model's accuracy decreases when a particular feature's values are randomly permuted. Studies such as [47,106] used these measures to rank hydrogeological parameters, consistently identifying slope, drainage density, lithology, and lineament density as the most influential factors. More recently, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) have gained prominence for their ability to provide both global and local interpretability. SHAP, grounded in cooperative game theory, assigns each feature a contribution value for every individual prediction, making it particularly valuable for understanding spatial heterogeneity in groundwater potential. For example, refs. [69,132] applied SHAP to interpret CatBoost and XGBoost models, demonstrating how slope and land use contributed differently across varying terrains. LIME, on the other hand, approximates the model locally to reveal the most influential parameters in

specific regions, offering fine-scale interpretive insights. Complementing feature importance measures, Partial Dependence Plots (PDPs) and Individual Conditional Expectation (ICE) plots visualize how changes in one or two predictors affect the predicted groundwater potential while holding others constant. These methods help identify nonlinear relationships and interactions among variables, for instance, how moderate slopes coupled with high lineament density increase groundwater occurrence probability [32]. Some studies also employ Accumulated Local Effects (ALE) plots to overcome extrapolation issues common in PDPs, providing more reliable interpretation in high-dimensional feature spaces [70]. Spatial interpretation tools, such as Geodetector, Geographical Detector Models, and Geographically Weighted Regression (GWR), have been incorporated to quantify spatial associations between groundwater occurrence and influencing factors. For instance, ref. [37] used the Geodetector method to measure the interaction effect (q-statistic) between lithology and lineament density, revealing that their combined influence exceeded individual effects. Such spatially explicit interpretation aligns ML findings with geological understanding, ensuring that model outputs adhere to hydrogeological logic. The growing integration of Explainable Artificial Intelligence (XAI) in GPM represents a paradigm shift from pure predictive modeling toward transparent, knowledge-informed learning. XAI frameworks enhance stakeholder trust by providing interpretable justifications for predictions, an essential aspect for groundwater management, where policy and planning decisions depend on model credibility. Future studies are expected to combine SHAP-based interpretability with uncertainty quantification, sensitivity analysis, and spatial validation, establishing explainable workflows that connect data science with hydrogeological reasoning. In summary, model interpretation and explainability serve dual purposes in GPM research: they verify the internal consistency of ML predictions and foster their acceptance among hydrogeologists and decision-makers. The increasing adoption of SHAP, PDPs, and Geodetector models marks a transition toward interpretable AI, promoting a deeper understanding of how environmental, geological, and climatic parameters interact to shape groundwater potential.

3.4.5. Sensitivity of Classification Schemes in Groundwater Zoning

The classification of continuous groundwater potential indices into discrete classes (e.g., very low, low, moderate, high, and very high) represents an important step in GPM because it directly affects the interpretation and usability of the resulting maps [17,18,41,138]. Various classification methods, such as natural breaks (Jenks), quantile, equal interval, and standard deviation, are commonly used in GIS-based studies [27,31,33]. However, previous research has shown that different classification schemes may produce substantially different spatial patterns and extents of groundwater potential zones, even when derived from the same model outputs [27–29]. This sensitivity arises because each classification method distributes data values differently across classes based on statistical characteristics of the dataset. Consequently, the selection of an appropriate classification method is essential for ensuring a realistic representation of groundwater potential zones and avoiding misinterpretation of model results. Several studies have highlighted that classification strategies should be chosen carefully according to the distribution of model outputs and the objectives of groundwater resource management, and the chosen method should be clearly justified in groundwater potential mapping studies [19,20,25].

A comparative summary of major machine learning model groups used in GPM is shown in Table A3. The comparative analysis indicates that tree-based ensemble models (G1) dominate recent GPM studies due to their robustness, strong predictive accuracy, and ability to handle nonlinear and multicollinear datasets. Hybrid and stacking frameworks (G6) are increasingly reported as top-performing approaches, reflecting a methodological

shift toward ensemble optimization. Regression-based models (G2) remain important for interpretability, whereas neural networks (G3) demonstrate strong potential in data-rich environments. Support vector machines (G4) provide stable performance under moderate dataset sizes but require careful hyperparameter tuning [40,64]. Bayesian approaches (G5), while less frequently used, offer advantages in uncertainty representation.

It can be noted that a distinct methodology evolution trend has been identified in GPM research in the last ten years. Initially, a number of machine learning models, such as logistic regression, support vector machines, and decision trees, were implemented. However, in recent years, a major trend towards the use of hybrid and ensemble machine learning models, such as random forests, gradient boosting, and XGBoost, as well as a combination of machine learning and metaheuristic models, has been identified. It has been noted that the use of ensemble machine learning models tends to improve the robustness and accuracy of groundwater potential modeling [40,64].

In recent years, a major trend towards the use of remote sensing-based conditioning factors in groundwater potential modeling has been identified. These conditioning factors include land use/land cover derived using satellite RS images, vegetation indices such as NDVI, lineament density, topographic wetness index, and climate reanalysis datasets. In addition, a major trend towards the use of feature selection and model interpretability has been identified. A number of machine learning models, such as recursive feature elimination and permutation feature selection in random forests and out-of-bag feature selection in ensemble machine learning models, have been implemented. In recent years, a major trend towards the use of explainable AI tools in groundwater potential modeling has been identified [40,64].

4. Key Challenges and Gaps in Current Research

Despite the significant progress achieved in applying ML for GPM, several methodological and practical challenges persist. These limitations stem from data quality issues, model selection uncertainties, computational constraints, and the limited interpretability of complex algorithms. Understanding these challenges is essential to improve the robustness, transferability, and real-world applicability of ML-based groundwater assessments.

4.1. Data Availability and Quality

The reliability of ML-based GPM largely depends on the quality, spatial resolution, and representativeness of input datasets. In many regions, especially in developing countries, the availability of consistent and high-resolution hydrogeological, soil, and climatic data remains limited [9]. Missing or incomplete borehole records, temporal inconsistencies in satellite imagery, and inaccuracies in geological or hydrological maps introduce uncertainty into the modeling process. Moreover, integrating multi-source data (e.g., DEMs, RS indices, and field data) often requires rigorous preprocessing to correct spatial misalignments, scaling differences, and projection errors. The scarcity of labeled data, such as groundwater well yields or recharge measurements, further constrains the development of reliable supervised learning models [11].

4.2. Feature Selection and Parameter Sensitivity

The selection of conditioning factors has a critical impact on model accuracy and interpretability. However, in practice, the inclusion of irrelevant or redundant variables can lead to overfitting and instability, while the exclusion of key hydrogeological parameters may cause underrepresentation of groundwater processes. Furthermore, the relationships between conditioning parameters (e.g., slope, lithology, drainage density) and groundwater occurrence are often nonlinear and site-dependent, making it difficult to generalize models

across different physiographic or climatic regions. Inconsistent use of feature selection techniques, such as correlation analysis, VIF, or mutual information, across studies leads to varied and sometimes contradictory results [16,96].

4.3. Model Uncertainty and Overfitting

ML models, particularly ensemble and deep learning approaches, are prone to overfitting when trained on small or imbalanced datasets. Overfitted models exhibit excellent performance on training data but perform poorly on unseen samples, limiting their generalization capability. The lack of standardized model validation procedures (e.g., k-fold cross-validation, spatial cross-validation) contributes to uncertainty in reported accuracies across studies. Moreover, stochastic elements in model training (e.g., random initialization, sampling bias) can lead to variability in outputs, especially in models such as Random Forest (RF), Gradient Boosting (GB), or Neural Networks (NNs). This variability underscores the need for uncertainty quantification and sensitivity analysis in GPM studies [139,140].

4.4. Computational Cost and Complexity

Advanced ML algorithms such as deep neural networks (DNNs), convolutional neural networks (CNNs), and hybrid ensemble models require substantial computational resources, particularly when dealing with high-dimensional spatial datasets. Hyperparameter optimization methods (e.g., grid search, Bayesian optimization, or genetic algorithms) significantly increase computational time and demand high-performance hardware. In addition, the increasing use of large-scale satellite and climate data (e.g., MODIS, CHIRPS) requires efficient data handling, preprocessing, and storage capabilities that may not be accessible to all research groups [80].

4.5. Model Transferability and Generalization

One of the persistent challenges in ML-based GPM is the transferability of models across different geographical regions. Models trained in one watershed or aquifer system may not perform well in another due to differences in hydrogeological, climatic, and land use conditions. This site-specific nature of model performance reduces the scalability and applicability of ML models for regional groundwater assessment. Domain adaptation techniques and transfer learning have been proposed as potential solutions, but their adoption in GPM remains limited. Ensuring model generalization thus requires careful calibration and validation using diverse datasets from multiple physiographic settings [11,81].

4.6. Lack of Explainability and Interpretability

While ML models such as Random Forest and XGBoost provide high predictive accuracy, they often operate as “black boxes”, offering limited insight into the physical or hydrogeological meaning of predictions. The lack of interpretability reduces trust and hinders the adoption of ML models by hydrogeologists and water resource managers. Recently, explainable artificial intelligence (XAI) methods such as SHAP, LIME, and Partial Dependence Plots (PDP) have been introduced to bridge this gap by quantifying variable importance and visualizing the influence of each parameter on the model output. However, these techniques are not yet widely standardized in GPM research and are sometimes interpreted incorrectly without sufficient hydrogeological justification [97].

4.7. Validation and Ground Truthing Limitations

Validation of GPM outputs often relies on limited ground truth data, typically spring or well locations, which may not comprehensively represent groundwater availability or productivity. Moreover, validation datasets are sometimes spatially autocorrelated with training samples, leading to overestimation of model performance. Field-based verification,

such as pumping tests, hydraulic head measurements, or tracer studies, remains rare in ML-based GPM studies due to logistical and financial constraints. Consequently, there is a pressing need for more comprehensive validation frameworks that integrate statistical, spatial, and field-based evaluation. In summary, while ML techniques have greatly enhanced the precision and automation of GPM, their broader adoption faces constraints related to data quality, computational efficiency, model interpretability, and cross-regional generalization. Future research should focus on establishing standardized protocols for data preprocessing, feature selection, and validation, as well as integrating physics-based and explainable ML approaches to ensure scientifically meaningful and operationally reliable GPM outputs [40,64,92,93].

5. Future Directions

ML has demonstrated strong potential in advancing GPM by improving predictive accuracy and integrating multi-source geospatial data. However, to transition from experimental research toward operational and decision-support tools for groundwater management, several future directions must be pursued. These directions address current methodological gaps and emerging technologies that can enhance the robustness, interpretability, and transferability of ML-based groundwater models.

5.1. Integration of Explainable Artificial Intelligence (XAI)

Future GPM studies should emphasize model transparency and interpretability through the integration of explainable artificial intelligence (XAI) techniques. Methods such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), Partial Dependence Plots (PDP), and Accumulated Local Effects (ALE) can provide quantitative insights into how input variables influence model predictions. These tools not only enhance scientific understanding of groundwater systems but also build confidence among policymakers and hydrogeologists by connecting ML outcomes with hydrogeological principles. Standardizing XAI frameworks across GPM studies will help improve model interpretability and support decision-making processes in groundwater management.

5.2. Development of Hybrid and Physics-Informed Models

One promising future direction is the integration of ML with physically based hydrological models, forming hybrid or physics-informed learning frameworks. While traditional ML models excel at pattern recognition, they lack explicit representation of groundwater flow and recharge dynamics. Combining ML with numerical models (e.g., MODFLOW, SWAT) or embedding physical constraints within the learning process can enhance model reliability and extrapolation capacity. For example, coupling Random Forest or XGBoost with Darcy's law-based simulation results can provide physically meaningful predictions that respect mass balance and aquifer properties. Such hybrid models represent a key step toward scientifically interpretable and policy-relevant GPM frameworks.

Recent advances in groundwater potential mapping research highlight several emerging directions that may significantly improve prediction reliability and practical applicability. One promising development is the integration of physics-informed machine learning, in which hydrological or hydrogeological constraints are incorporated into data-driven models to improve their physical consistency and interpretability. Another important research direction involves uncertainty quantification, which aims to evaluate the reliability and robustness of model predictions by analyzing the influence of data quality, model parameters, and algorithmic assumptions. In addition, the transferability of groundwater potential models across different hydrogeological regions remains an important challenge,

as models developed for a specific geological setting may not perform equally well in other environments due to variations in lithology, climate, and groundwater recharge processes. Furthermore, the increasing availability of multi-source remote sensing data, including optical imagery, radar observations, and digital elevation model (DEM)-derived parameters, has created new opportunities for improving the spatial representation of groundwater conditioning factors. These emerging directions are expected to play an important role in advancing the accuracy, robustness, and applicability of future groundwater potential mapping studies [141].

5.3. Transfer Learning and Domain Adaptation

Current ML-based GPM models often lack transferability across regions due to differences in hydrogeological settings, climate, and data availability. The adoption of transfer learning and domain adaptation techniques can enable pre-trained models to be fine-tuned for new regions using limited local data. This approach reduces the dependency on large labeled datasets and facilitates the reuse of trained models across watersheds with similar hydrogeological features. Such advances are crucial for scaling ML applications to data-scarce regions, particularly in developing countries where field-based information is limited.

5.4. Incorporation of Temporal Dynamics

Most existing GPM studies adopt a static approach, treating groundwater potential as a time-invariant property. However, groundwater systems are inherently dynamic and influenced by seasonal rainfall, land use change, and anthropogenic extraction. Future research should therefore incorporate temporal variability by integrating multi-temporal satellite datasets (e.g., MODIS, Sentinel, GRACE) and time-series ML models such as Long Short-Term Memory (LSTM) or Temporal Convolutional Networks (TCNs). These models can capture the temporal evolution of groundwater systems, allowing for near-real-time prediction of recharge or depletion zones under changing climatic and anthropogenic conditions.

5.5. Advances in Data Fusion and Big Data Analytics

With the rapid growth of Earth observation and IoT-based monitoring systems, future GPM research can leverage data fusion and big data analytics to enhance model accuracy and scalability. Integrating multi-resolution datasets, such as LiDAR-derived topography, hyperspectral imagery, GRACE-based groundwater storage anomalies, and in situ hydrochemical measurements, can improve the spatial representation of groundwater-related processes. Techniques such as autoencoders, deep feature extraction, and graph-based learning offer promising avenues for handling large, heterogeneous datasets in GPM. The use of cloud-based geospatial platforms like Google Earth Engine (GEE) and HydroDataCube will further streamline large-scale model implementation and data processing.

5.6. Uncertainty Quantification and Sensitivity Analysis Future

GPM frameworks should systematically address model uncertainty and sensitivity to strengthen the credibility of ML-based predictions. Bayesian inference, Monte Carlo simulation, and ensemble learning can be employed to quantify prediction uncertainty arising from data noise, model bias, or parameter variability. Sensitivity analysis methods such as Sobol indices or Morris screening can identify the most influential parameters affecting groundwater potential. These methods enhance both scientific understanding and practical decision-making by quantifying the confidence associated with each prediction.

5.7. Spatial Cross-Validation and Field-Based Evaluation

Traditional random cross-validation approaches often overlook spatial autocorrelation, leading to overoptimistic model performance estimates. Therefore, spatial cross-validation (SCV) should be adopted as a spatially explicit statistical validation strategy that provides more realistic and unbiased accuracy assessments for spatial datasets. In addition to these statistical validation approaches (including both random and spatial cross-validation), greater emphasis should be placed on independent field-based evaluation, which involves comparing model predictions with external hydrogeological observations not used during model development. Such evaluations may include hydraulic head measurements, pumping test data, and isotope tracing, providing direct evidence of groundwater conditions.

It is important to distinguish these independent assessments from datasets used for model training, validation, and testing. Accordingly, a comprehensive evaluation framework should integrate statistical validation approaches (random and spatial cross-validation) with independent field-based evaluation. This combined strategy enhances both the reliability and the practical applicability of machine learning-based groundwater potential mapping.

5.8. Open-Source Frameworks and Reproducibility

To facilitate broader scientific collaboration, future GPM studies should prioritize open data, open models, and reproducible workflows. Establishing shared repositories for datasets (e.g., borehole records, DEMs, LULC layers) and model codes (e.g., through GitHub or HydroShare) will promote transparency and accelerate methodological innovation. Integration with open-source ML libraries (e.g., Scikit-learn, TensorFlow, CatBoost, LightGBM) and reproducible pipelines (e.g., MLflow, DVC) will also improve comparability across studies and reduce redundancy in model development.

5.9. Towards Decision-Support Systems and Sustainable Groundwater Management

Finally, the ultimate goal of ML-based GPM should extend beyond academic model development to practical decision-support systems (DSSs) that aid policymakers and water managers. Future frameworks should integrate socio-economic and climate change factors to predict groundwater sustainability under different management scenarios. Coupling ML-based GPM with GIS-based multi-criteria decision analysis (MCDA) or groundwater vulnerability mapping can provide actionable insights for aquifer protection, well placement, and water allocation strategies. In summary, the future of ML-based GPM lies in the convergence of explainable AI, hybrid physical–data-driven modeling, temporal analysis, and reproducible open-science practices. By incorporating uncertainty quantification, domain adaptation, and multi-source data fusion, next-generation models can transition from predictive tools to reliable, interpretable systems that support sustainable groundwater resource management worldwide.

6. Conclusions

This extensive review has critically assessed the present status of groundwater potential mapping (GPM) through machine learning (ML) approaches, integrating the evolution of methodologies, performance trends of models, feature selection methods, validation procedures, and future research trends. Based on the analysis of 83 selected studies, the present trend of methodologies is shifting from traditional statistical models to more sophisticated ML-based and ensemble models.

1. The trend is towards the increasing use of tree-based and ensemble models, especially Random Forest (RF), Gradient Boosting, and ensemble models, owing to their ability to handle multicollinearity, model nonlinear relationships, and make accurate predic-

tions. Although traditional statistical models like Frequency Ratio (FR) and Logistic Regression (LR) are still useful for their interpretability and baseline comparison, ensemble models based on ML consistently outperform other models in terms of accuracy and generalization capacity. Additionally, the trend is towards the use of hybrid models to combine the complementary strengths of different models.

2. The review confirms that some conditioning variables are consistently identified as the most important predictors of groundwater potential in different climatic and geological regions. Variables such as slope, lithology, drainage density, lineament density, elevation, topographic wetness index (TWI), land use/land cover (LULC), and rainfall are some of the most important variables that contribute to the prediction of groundwater potential. The increasing use of high-resolution remote sensing (RS) datasets and DEM-derived variables has greatly improved the spatial representation of hydrogeological variables.
3. Feature selection and validation methods have evolved significantly. There is an increasing use of multicollinearity checks (VIF, Tol), entropy-weighting, mutual information, recursive feature elimination, and ensemble-based out-of-bag (OOB) importance estimation to enhance the relevance of predictors. Validation methods have also moved towards more robust approaches, such as k-fold cross-validation, spatial validation approaches, ROC-AUC analysis, and multi-metric validation. However, issues such as spatial autocorrelation bias and the lack of external validation remain as challenges.

Despite these developments, there are still some limitations that continue to influence GPM studies. These include data unavailability in arid and semi-arid environments, lack of documentation of hyperparameter optimization methods, poor uncertainty estimation, and a lack of focus on explainability. While predictive accuracy has been enhanced, interpretability and hydrogeological consistency of ML models still require emphasis to ensure hydrogeological reliability.

Future studies should focus on the following:

1. The application of explainable AI methods (such as SHAP and permutation importance) to improve model interpretability.
2. The inclusion of uncertainty analysis and probabilistic mapping.
3. The development of hybrid deep learning-ensemble models for complex terrain.
4. The utilization of high-resolution multi-source remote sensing and climate re-analysis data;
5. The development of standardized benchmarking datasets for fair cross-model comparison.

In conclusion, ML-based GPM has evolved from exploratory modeling to a more structured, optimization-oriented, and validation-conscious field of research. Further convergence of innovative algorithms, explainability tools, and hydrogeological knowledge will be critical for developing robust, transferable, and decision-support-oriented GPM models to meet the growing demands on water resources under climate variability and human-induced stress.

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Appendix A

Table A1. Data density derived from previous machine learning-based groundwater potential mapping studies, showing the relationship between the study area and the number of groundwater inventory points used for model development.

Study Area (Area, km ²)	Total Number of Inventory Data Points	Data Density, Points/km ²	Reference
Iraqi southern desert, Iraqi (78,390 km ²)	93 perennial flowing wells	0.001	[43]
Damghan sedimentary plain, Iran (1559 km ²)	80 wells	0.051	[49]
Shahrud plain, Iran (665.36 km ²)	122 wells	0.183	[47]
Yasuj-Dena area, Iran (2159.9 km ²)	362 spring locations	0.168	[142]
Lailan basin, Iraq (422 km ²)	1031 operating wells	2.443	[68]
Lake Urmia basin, Iran (51,826 km ²)	10,624 springs	0.205	[72]
Kabgian watershed, Iran (873 km ²)	384 springs	0.440	[143]
Nizwa watershed, Oman	336 points (168 aflaj, 168 non-aflaj)	Five classes: 0.60–3.31, 0.31–4.42, 4.42–5.49, 5.49–6.79, and 6.79–9.14 km/km ²	[94]
Boroujen, Iran (580 km ²)	429 locations	0.740	[99]
Guelma basin, Algeria (841.5 km ²)	152 water points	0.181	[100]
Gandheswari Watershed, India (394.96 km ²)	170 sites (85 wells, 85 non-wells)	0.430	[44]
Azraq Basin, Jordan (7200 km ²)	464 wells	0.064	[75]
Wadi Yalamlam, Saudi Arabia (404.3 km ²)	168 points	0.416	[51]
Toudgha Oasis, Morocco (2296 km ²)	884 binary points	0.385	[17]
Bankura district, India (6882 km ²)	231 points	0.034	[84]
Lijiang River Basin, China (5815.91 km ²)	356 points	0.061	[69]
Fars province, Iran (123,144 km ²)	1312 piezometric wells	0.011	[11]

Table A1. *Cont.*

Study Area (Area, km ²)	Total Number of Inventory Data Points	Data Density, Points/km ²	Reference
KUC, West Bengal, India (3607.94 km ²)	312 points (156 wells, 156 non-wells)	0.086	[28]
Hubei Region, China (185,900 km ²)	812 samples (406 groundwater, 406 non-groundwater)	0.004	[37]
Central Highlands, Vietnam (54,700 km ²)	501 wells	0.009	[57]
Teesta sub-catchment, Bangladesh (2284 km ²)	220 well points	0.096	[48]
Saveh City, Iran (4931 km ²)	349 spring points	0.071	[93]
Leigh Creek, South Australia (1206 km ²)	64 presence locations (wells > 5 L/s)	0.053	[27]
Narayanganj, Bangladesh (760 km ²)	349 well and non-well locations	0.459	[98]
Idukki district, India (4358 km ²)	253 well locations (121 low, 132 high yield)	0.058	[55]

Notes: Inventory points refer to observed groundwater occurrence data (e.g., wells, springs, or boreholes) used as presence samples for model training, validation, and testing. *Non-points* (or non-groundwater locations) refer to absence or pseudo-absence samples used in classification-based models to represent locations without confirmed groundwater occurrence. Data density (points/km²) is calculated as the ratio between the total number of inventory points and the study area. The number and distribution of presence and absence samples vary across studies, depending on data availability and sampling strategy. In many groundwater potential mapping studies, reliable absence data are often unavailable, as the lack of observed groundwater at a location does not necessarily indicate true absence. To address this limitation, some recent studies have adopted Positive–Unlabeled (PU) learning, in which only confirmed groundwater occurrence data (presence samples) are used. In contrast, the remaining data are treated as unlabeled rather than a true absence. This approach reduces the risk of introducing bias associated with incorrectly labeled absence data and has shown promising results in improving model robustness, particularly in data-scarce regions. Although PU learning is still relatively underrepresented in the GPM literature, it represents a promising direction for future research.

Table A2. Summary of representative studies illustrating the methodological aspects of ML model development and validation techniques for groundwater potential mapping between 2015 and 2025.

Data Size	Division, % (Training/Testing)	Tuning	Evaluation	Explainability	References
1425	70/30	N/A	SRC, PRC, AUC	N/A	[40]
864	70/30	N/A	ROC, AUC	N/A	[42]
128	70/30	N/A	ROC, AUC	N/A	[43]
1224	70/30	N/A	ROC, AUC	N/A	[45]
122	70/30	N/A	AUC, SCAI	N/A	[47]
85	70/30	BO	CMA, ROC, CV	EIA	[49]
80	70/30	N/A	AUROC, SS, ACC, MAE, RMSE, SCAI	N/A	[50]
917	70/30	N/A	ROC, AUC	N/A	[52]
66	70/30	N/A	AUC, CI, SE, FT, WT	N/A	[54]
N/A	N/A	N/A	AUC-ROC, TSS, E, FPR, TPR, TNR, FNR	N/A	[56]
167	70/30	N/A	AUC-ROC	N/A	[58]
227	70/30	N/A	AUC, Kappa, RMSE, SS, ACC, PPV, NPV	N/A	[60]
53	70/30	N/A	ROC, AUC	N/A	[61]

Table A2. Cont.

Data Size	Division, % (Training/Testing)	Tuning	Evaluation	Explainability	References
141	70/30	N/A	AUC, TSS	EIA	[63]
72	70/30	N/A	AUC, RMSE, Kappa, ACC, SS, NPV, PPV	N/A	[65]
506	70/30	GSr	AUC, Kappa, ACC, PPV, NPV, SS	EIA	[66]
227	70/30	N/A	AUC, NPV, SST, RMSE, ACC, SPF, PPV, Kappa	N/A	[68]
339	70/30	N/A	ACC, Kappa, Precision, Recall	N/A	[69]
1031	70/30	GSr	ACC, Kappa	N/A	[66]
10	75/25	N/A	AUC-ROC, ACC, SS	EIA	[72]
185	70/30	N/A	AUC, ACC, Precision, Recall, F1	N/A	[14]
374	75/25	RSr	ACC, SS, Precision, FPR, MCC, Kappa, AUC, RMSE, MAE	N/A	[61]
84	70/30	GSr	ACC, SS, PPV, NPV, AUC-ROC	N/A	[74]
227	70/30	N/A	AUC, RMSE, Kappa, ACC, SPF, SST, NPV, PPV	N/A	[76]
187	70/30	N/A	RMSE, MAE, Kappa, Precision, Recall, SS, ROC-AUC	N/A	[78]
484	N/A	N/A	ROC-AUC	N/A	[79]
512	70/30	N/A	ACC, Precision, Recall, F-score, AUC-ROC	N/A	[81]
2884	N/A	N/A	ROC, AUC	N/A	[83]
10	75/25	GSr	AUC-ROC, FT, WT	N/A	[32]
714	70/30	N/A	AUROC, TSS	EIA	[85]
1598	N/A	N/A	AUROC	N/A	[87]
113	80/20	RSr	R, RMSE, MABE, NSC	N/A	[88]
488	70/30	RSr	Test Score, Balanced Score, AUC	PDP	[90]
483	70/30	N/A	Balanced ACC, G-Mean, AUC	PDP	[91]
349	70/30	N/A	SS, PPV, NPV, AUC-ROC	N/A	[93]
336	70/30	GSr, RSr, BO	AUC	N/A	[94]
50	N/A	RSr	AUC-ROC	N/A	[21]
75	70/30	GSr	AUC, ACC, SS, RMSE	N/A	[97]
429	70/30	N/A	AUC-ROC, RMSE	EIA	[99]
152	70/30	N/A	ROC, AUC, ERT	N/A	[100]
100	N/A	N/A	ROC, AUC	EIA	[101]

Table A2. Cont.

Data Size	Division, % (Training/Testing)	Tuning	Evaluation	Explainability	References
460	80/20	N/A	eROC, bROC	EIA	[18]
2513	70/30	GSr	Precision, Recall, SS, ACC, F1, ROC, AUC	EIA	[41]
10	75/25	GSr	AUC-ROC, FT, WT, PPV, NPV, SST, SPF, ACC	N/A	[32]
170	N/A	N/A	ACC, Kappa, RMSE, MAE, R ²	N/A	[44]
3345	N/A	RSr	F1, AUC	EIA	[46]
N/A	N/A	N/A	Comparison with Existing Dams	N/A	[144]
220	80/20	N/A	ROC, AUC	EIA	[48]
812	80/20	N/A	ROC-AUC, Precision, Recall, F1	SHAP	[37]
168	70/30	N/A	ACC, SS, F1	EIA	[51]
85	N/A	N/A	ACC	N/A	[53]
253	70/30	N/A	AUC, ACC, Kappa	N/A	[55]
501	70/30	N/A	ACC, PPV, NPV, SST, SPF, AUC, RMSE	N/A	[57]
1371	N/A	N/A	ROC, AUC	N/A	[59]
351	70/30	GSr	R ² , RMSE, MAE, AUC	EIA	[39]
80	80/20	N/A	AUC	N/A	[121]
100	70/30	GSr	ACC, SS, F1, MCC	EIA	[62]
59	70/30	N/A	ROC-AUC, RMSE, MAE	EIA	[64]
2	N/A	RSr	ROC, ACC, Precision, Recall, F1, Kappa	SHAP	[16]
N/A	N/A	N/A	AUC-ROC, RMSE, MAE	N/A	[67]
2	70/30	N/A	ROC, AUC, ACC, SS	N/A	[31]
800	N/A	BO	ACC, Precision, AUC, Recall, F1	N/A	[70]
127	N/A	N/A	ACC, AUC, RMSE, Kappa	N/A	[71]
174	55/45, 65/35, and 75/25	GSr	MSE, RMSE, R ² , F1, Kappa, MCC, SS, Precision, FPR, ACC, AUC	N/A	[73]
N/A	70/30	N/A	ROC, Correlation	N/A	[36]
884	N/A	N/A	SS, Precision, FPR, ACC, F1, MAE, RMSE, AUC-ROC	N/A	[17]
464	70/30	N/A	MSE, RMSE, AUROC	N/A	[75]
101	70/30	N/A	PPV, NPV, SS, ACC, RMSE, AUC	N/A	[77]
200	N/A	N/A	ROC, AUC	N/A	[20]

Table A2. Cont.

Data Size	Division, % (Training/Testing)	Tuning	Evaluation	Explainability	References
75	70/30	N/A	CMa, ACC, Kappa, AUC-ROC	N/A	[80]
146	70/30	RSr	AUC, RMSE, R ² , MAE	N/A	[82]
501	70/30	N/A	ROC, ACC, NPV, PPV, RMSE, SS, Kappa, AUC	N/A	[33]
161	70/30	GSr	Precision, Recall, F1, ACC, ROC-AUC	SHAP	[84]
195	N/A	N/A	AUC, RMSE, R ² , ACC, Kappa	N/A	[86]
39	70/30	GSr	AUC-ROC, ACC, F1	N/A	[29]
N/A	75/25	N/A	R ² , NASH, PBIAS, RMSE	N/A	[89]
356	70/30	N/A	MSE, RMSE, ACC, F-score, SRC, PRC	SHAP	[69]
601	N/A	DBO	Raincloud Plots, AUC, ACC, TPR, TNR	EIA	[92]
46	70/30	GSr	ACC, RMSE, R ²	EIA	[30]
N/A	N/A	N/A	ROC, AUC, ACC, Precision, Recall, F1	N/A	[95]
N/A	N/A	N/A	ACC, Precision, Recall, F1, CMa	EIA	[96]
349	70/30	N/A	ROC, AUC-ROC	N/A	[98]
N/A	70/30	GSr, RSr	R ² , RMSE	EIA	[12]
312	70/30	N/A	ROC-AUC	N/A	[28]
64	70/30	N/A	ROC, AUC, R ² , MAE, RMSE, SE	N/A	[27]

Notes: EIA = Feature importance analysis, BO = Bayesian Optimization, GSr = Grid Search, RSr= Random search, DBO = Dung Beetle Optimizer, SHAP = Shapley Additive Explanations, PDP = Partial Dependence Plot, ACC: Accuracy, AUC = Area Under the Curve, AUC-ROC = Area Under the Receiver Operating Characteristic Curve, AUROC = Area Under the Receiver Operating Characteristic Curve, bROC = Binormal Receiver Operating Characteristic, CI = Confidence Interval, E = Efficiency, eROC = Empirical Receiver Operating Characteristic, F1 = F1-Score, FNR = False Negative Rate, FPR = False Positive Rate, MABE = Mean Absolute Bias Error, MAE = Mean Absolute Error, MCC = Matthews Correlation Coefficient, MSE = Mean Squared Error, NASH = Nash–Sutcliffe Coefficient, NPV = Negative Predictive Value, NSC = Nash–Sutcliffe Coefficient, PBIAS = Percent Bias, PPV = Positive Predictive Value, PRC = Prediction Rate Curve, SS= Specificity and Sensitivity.

Table A3. Comparative summary of major machine learning model groups used in groundwater potential mapping (GPM).

Model Group	Predictive Performance (General Trend)	Data Requirements	Strengths	Limitations	Suitable Applicability Conditions
G1: Tree/Ensemble Models	High to very high; consistently among top-performing models in most GPM studies	Moderate to large datasets; handle high-dimensional data well	Robust to multicollinearity; captures nonlinear relationships; handles mixed data types; intrinsic feature importance (e.g., OOB)	Higher computational cost (boosting); may require tuning; less transparent than linear models	Complex hydrogeological settings; nonlinear predictor–response relationships; datasets with correlated variables

Table A3. Cont.

Model Group	Predictive Performance (General Trend)	Data Requirements	Strengths	Limitations	Suitable Applicability Conditions
G2: Regression-Based Models	Moderate; effective as baseline models	Moderate dataset size; requires relatively clean and well-structured data	High interpretability; low computational cost; clear statistical inference	Limited ability to model nonlinear relationships; sensitive to multicollinearity	Preliminary assessments; studies prioritizing interpretability and statistical inference
G3: Neural Networks	High when sufficient data available	Large datasets preferred; sensitive to data scaling and noise	Strong nonlinear modeling capability; flexible architecture	Risk of overfitting; high computational demand; low interpretability; requires careful tuning	Data-rich environments; complex feature interactions; high-resolution spatial datasets
G4: Support Vector Machines	High in medium-sized datasets; stable performance	Moderate dataset size; sensitive to parameter selection (C, γ)	Effective in high-dimensional space; good generalization; robust to overfitting	Parameter tuning critical; computationally intensive for large datasets	Medium-sized datasets; complex but structured predictor spaces
G5: Bayesian Models	Moderate; context-dependent	Small to moderate datasets; probabilistic input structure	Probabilistic interpretation; handles uncertainty explicitly	Assumptions of independence (NB); structure learning complexity (BN)	Studies emphasizing uncertainty modeling and probabilistic mapping
G6: Hybrid/Ensemble Combinations	Very high; often outperform single models	Moderate to large datasets; increased computational demand	Combines the strengths of multiple models; improved robustness and accuracy	Increased complexity; reduced interpretability; higher computational cost	Advanced research applications; regional/national-scale mapping; studies seeking optimal predictive performance

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